Decentralized Crypto World Bank

A Blockchain-Based Lending Platform
with AI-Enhanced Security

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

Department of Computer Science and Engineering
Brac University
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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at Brac University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Approval

The pre-thesis 1 report of final year project titled "Decentralized Crypto World Bank: A Blockchain-Based Lending Platform with AI-Enhanced Security" submitted by

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Of Spring, 2026 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science on February 20, 2026.

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Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers.

Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a prototype framework that models a multi-tier institutional lending architecture on a programmable blockchain infrastructure. We show that hierarchical capital flows, role-based governance, and data-informed risk analytics can be coordinated within a unified smart contract environment, enabling transparent state visibility alongside adaptive decision support. This hybrid design illustrates how institutional finance and decentralized systems may converge, suggesting a pathway toward more accountable, flexible, and inclusive digital credit ecosystems.

Keywords: Blockchain, Decentralized Finance, Institutional Architecture, Financial Inclusion, Smart Contracts, AI-Augmented Governance

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

Acknowledgment

We would like to express our sincere gratitude to our supervisor, Mr. Annajiat Alim Rasel, Senior Lecturer at the Department of Computer Science and Engineering, Brac University, for his guidance and support throughout this project. We also thank the Department of Computer Science and Engineering at Brac University for providing us with the resources and academic environment to pursue this work.

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Chapter 1

Introduction

The convergence of blockchain technology, decentralized finance, and artificial intelligence presents new opportunities to address longstanding inefficiencies in global development finance. This project, the Crypto World Bank, explores how a decentralized lending platform with a multi-tier institutional structure can improve transparency, reduce settlement friction, and expand access to credit in underserved markets. The complete source code, smart contracts, and supplementary documentation for this project are available in our GitHub repository.

We position blockchain in this context as a coordination technology that provides shared state visibility, tamper-evident audit trails, and programmable process enforcement across multiple institutions. The platform further addresses governance design, security controls, and regulatory considerations through a phased deployment pathway involving academic validation and regulatory sandbox pilots. By integrating a four-tier institutional hierarchy with transparent ledger infrastructure and data-driven risk analytics, the Crypto World Bank explores a hybrid architecture for institutional decentralized finance that targets improved operational efficiency and expanded access to credit in emerging markets.

1.1 Background

International development finance is delivered through layered institutional structures in which multilateral development institutions (e.g., World Bank, International Monetary Fund), national financial intermediaries, and local lenders jointly channel capital to end borrowers. While this model enables risk distribution and local market access, it is associated with operational challenges: limited cross-tier transparency, lengthy cross-border settlement processes, high reconciliation and compliance overhead, and constrained scalability of risk assessment workflows.

Simultaneously, approximately 1.4 billion adults remain unbanked globally according to the World Bank Global Findex [14], highlighting the need for more efficient and accessible credit delivery mechanisms. The DeFi ecosystem has demonstrated that smart contract platforms can automate lending with full transparency; however, existing DeFi protocols employ flat, pool-based architectures that do not model institutional hierarchy or integrate AI-driven risk analytics.

1.2 Rationale of the Study

This study is motivated by the convergence of three factors: (1) the persistent inefficiencies in traditional development finance that limit transparency and settlement speed; (2) the gap between DeFi's technical capabilities and its failure to address multi-tier institutional lending use cases; and (3) the opportunity to combine blockchain's auditability with AI/ML security analytics to improve both operational efficiency and decision quality. By designing a prototype that bridges these domains, we aim to demonstrate a viable path toward transparent, programmable, and AI-augmented lending for underserved populations.

1.3 Problem Statement

The global development finance ecosystem is characterized by a hierarchical topology in which supranational entities (e.g., the World Bank, International Monetary Fund) allocate capital to sovereign national institutions, which in turn disburse funds to regional and local banking entities, ultimately reaching end-user borrowers. This multi-layered intermediation introduces several systemic inefficiencies:

1. **Opacity of ledger state.** Internal bookkeeping across tiers is not publicly auditable. Stakeholders at lower tiers have limited visibility into reserve adequacy, approval criteria, and fund utilization at higher tiers.
2. **Settlement latency.** Cross-border capital transfers between tiers involve multiple intermediaries, compliance checkpoints, and reconciliation processes, resulting in settlement periods ranging from days to weeks.
3. **Manual risk assessment.** Creditworthiness evaluation and fraud detection are predominantly human-driven, introducing inconsistency, bias, and scalability constraints.
4. **Trust deficit.** The effort required to establish inter-institutional trust through legal agreements, audit engagements, and compliance certifications constitutes a significant overhead that disproportionately affects smaller institutions and borrowers in developing economies.

Simultaneously, the DeFi ecosystem has demonstrated that smart contract platforms can automate lending, collateral management, and interest computation with full transparency. However, existing DeFi protocols (e.g., Aave, Compound, MakerDAO) exhibit fundamental limitations in this context:

- They employ **peer-to-pool** or **over-collateralized** lending models that do not accommodate institutional hierarchies.

- They lack integration with **AI-driven risk analytics** and **explainable decision support** systems.
- They do not model **role-based, tiered governance** analogous to development finance institutions.

1.4 Objectives

The objectives of this project are:

1. To design and implement a four-tier lending architecture (World Bank → National Bank → Local Bank → Borrower) on an EVM-compatible blockchain that preserves institutional hierarchy while enabling shared ledger visibility.
2. To integrate off-chain AI/ML security analytics—including fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP)—to augment human credit governance.
3. To demonstrate transparent, programmable lending workflows with configurable borrowing limits, installment schedules, and role-based access control enforced by smart contracts.
4. To validate the prototype on public testnets (Polygon, Ethereum Sepolia) with functional verification of hierarchical controls, transparency properties, and operational workflows.
5. To document governance design, security controls, regulatory considerations, and a phased deployment pathway for academic validation and potential regulatory sandbox pilots.

1.5 Blockchain Justification

The problem described above is fundamentally a **multi-party coordination and trust challenge** involving disparate institutions with potentially misaligned incentives. We assert that blockchain technology addresses this class of problems more effectively than any conventional alternative for the following reasons:

1. **Trust minimization through consensus.** Distributed consensus mechanisms ensure that no single party can unilaterally alter the ledger state, eliminating the need for a trusted central operator.
2. **Programmable enforcement.** Smart contracts codify lending rules—borrowing limits, installment schedules, approval workflows—as deterministic, self-executing programs, reducing reliance on manual processes and bilateral agreements.
3. **Cryptographic auditability.** All state transitions are recorded in an immutable, publicly verifiable transaction log, enabling real-time auditing by regulators, partners, and borrowers without requiring costly third-party engagements.

4. **Composable incentive structures.** On-chain reputation systems, governance tokens, and programmable fee distributions can align incentives across all network participants.

A conventional cloud-hosted database infrastructure would necessitate a trusted central operator to maintain ledger integrity, enforce access control, and provide audit guarantees—reintroducing the very trust dependencies that this project seeks to mitigate. Blockchain eliminates this single point of trust failure.

1.6 Proposed Solution

The Crypto World Bank implements a **four-tier, on-chain lending architecture**:

- **Tier 1 — World Bank:** Maintains the global crypto reserve. Allocates capital to registered National Banks through smart contract-mediated lending operations.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to registered Local Banks within their jurisdiction. Aggregate risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrow from National Banks. Process loan requests from individual borrowers. Employ designated approvers for loan lifecycle management.
- **Tier 4 — Borrowers:** Submit loan requests to Local Banks. Repay through configurable installment schedules. Accumulate on-chain repayment history that informs future borrowing limits.

The platform further integrates:

- **Risk-aware borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

1.7 Methodology in Brief

We adopted a lightweight Agile/Scrum methodology with three sprints over an 8-week development window. Sprint 1 establishes smart contracts, frontend scaffolding, wallet integration, and database schema. Sprint 2 delivers lending features (loan requests, approvals, installments, borrowing limits), chat system, income verification, and bank registration. Sprint 3 integrates AI/ML fraud detection, risk dashboard, security audit, and documentation. Development uses Solidity 0.8.20 for smart contracts, React 18 with TypeScript for the frontend, FastAPI for the backend, and scikit-learn for ML models. We deploy on Polygon and Ethereum testnets at zero real-cryptocurrency cost.

1.8 Scopes and Challenges

Scope. Our prototype operates exclusively on public testnets (no real cryptocurrency). In-scope features include: four-tier lending, loan request and approval workflow, installment payments, borrowing limit computation, borrower–bank chat, income verification, fraud detection (Random Forest + SHAP), and anomaly detection (Isolation Forest). The dual-currency facility is designed as an auxiliary add-on to existing banking infrastructure. Out of scope for the prototype: mainnet deployment, fiat on-ramp/off-ramp integration, KYC/AML compliance automation, and production-scale stress testing.

Challenges. Key challenges include: (1) *Data scarcity*—labeled fraud data for DeFi lending is limited; synthetic or transfer learning may be required. (2) *Regulatory uncertainty*—crypto lending faces jurisdictional restrictions; our prototype mitigates this by operating on testnets only. (3) *Gas cost sensitivity*—complex on-chain logic increases transaction cost; the design partitions AI/ML and heavy computation off-chain. (4) *Model interpretability*—regulatory requirements for explainable lending decisions are addressed via SHAP-based feature attribution.

Chapter 2

Literature Review

2.1 Preliminaries

This section establishes key terminology and concepts used throughout the literature review and the project design.

Decentralized Finance (DeFi). DeFi refers to financial services built on blockchain platforms that operate without traditional intermediaries. Lending protocols enable users to borrow and lend assets through smart contracts, with interest rates typically determined algorithmically or by market utilization.

Smart contracts. Self-executing programs deployed on a blockchain that execute when predefined conditions are met. In lending, smart contracts handle deposits, loan requests, approvals, repayments, and interest computation without human intermediaries.

Over-collateralization. A lending model where borrowers must pledge collateral worth more than the loan amount. This is the dominant model in DeFi (e.g., Aave, Compound) due to the absence of credit scoring; it excludes under-collateralized lending common in traditional finance.

Explainable AI (XAI). Techniques that make machine learning model predictions interpretable to humans. SHAP and Local Interpretable Model-agnostic Explanations (LIME) are prominent methods for attributing feature importance to individual predictions, critical for regulatory compliance in lending decisions.

Proof of Stake (PoS). A consensus mechanism where validators secure the network by staking tokens; used by Ethereum (post-Merge) and Polygon. PoS is more energy-efficient than Proof of Work and enables faster block finality.

Central Bank Digital Currency (CBDC). A digital form of a country's fiat currency issued by the central bank. CBDCs use tiered distribution models that parallel the multi-tier architecture explored in this project.

2.2 Review of Existing Research

The design of the Crypto World Bank draws on a focused review of seven peer-reviewed publications spanning five domains directly relevant to our architecture.

2.2.1 Decentralized Lending and DeFi Protocol Design

Werner et al. [1] systematize DeFi protocol design in their SoK paper, identifying that existing lending platforms are uniformly pool-based and over-collateralized, lacking institutional hierarchy—a gap this project directly addresses. Their taxonomy of DeFi protocols reveals that no existing system models multi-tier capital flow analogous to development finance, confirming the novelty of our four-tier approach.

Bastankhah et al. [2] introduce an adaptive, data-driven DeFi lending protocol with a dual fast/slow control architecture, demonstrating that dynamic interest rate adjustment significantly outperforms static utilization curves used by Aave and Compound. Their work validates the feasibility of algorithmic lending parameter optimization, an approach we extend through our configurable interest rate tiers across the four-level hierarchy.

2.2.2 Machine Learning for Blockchain Security

Palaiokrassas et al. [3] apply machine learning to multichain DeFi fraud detection across 23 protocols encompassing over 54 million transactions, demonstrating that DeFi-specific behavioral features improve XGBoost and Neural Network classifiers to F1-scores of 0.76–0.85, compared to 0.08 with transactional features alone. Their finding that behavioral features outperform raw transaction data directly informed our feature engineering approach for the Random Forest fraud detection model.

Liu et al. [6] introduce the Isolation Forest algorithm for unsupervised anomaly detection. The algorithm isolates anomalies by recursively partitioning data, assigning shorter path lengths to outliers. We adopted Isolation Forest as our secondary detection model for wallet behavior analysis where labeled fraud data is scarce, enabling detection of novel attack patterns without pre-labeled training data.

2.2.3 Explainable AI in Financial Decision Systems

Adom et al. [4] provide a direct comparison of LIME and SHAP on loan approval systems, finding that SHAP provides deeper, more consistent feature attributions via Shapley values, while LIME offers faster runtime but less stable explanations across repeated evaluations. Their comparative analysis informed our decision to adopt SHAP as the primary explainability method for loan risk assessments, balancing interpretability depth with regulatory compliance requirements.

2.2.4 Blockchain for Financial Inclusion

Tan [5] develops an International Monetary Fund (IMF) model showing CBDCs in developing countries can bank large unbanked populations through a two-tier distribution model (central bank → commercial banks → users) that directly parallels our four-tier hierarchy. The study demonstrates that digital currency infrastructure, when distributed through existing institutional channels, can reach populations excluded from traditional banking—supporting the Crypto World Bank's mission of transparent, programmable lending for underserved populations.

2.2.5 Smart Contract Security and Governance

Atzei et al. [7] catalog Ethereum smart contract attack vectors including reentrancy, integer overflow, and access control vulnerabilities. Their taxonomy directly informed our use of OpenZeppelin's ReentrancyGuard, Solidity 0.8.20 built-in overflow protection, and role-based access control modifiers. We adopted their recommended defense-in-depth strategy of combining security primitives with planned formal verification using static analysis tools (Slither) and symbolic execution (Mythril).

2.3 Literature Review Summary

Table 2.1 summarizes the seven reviewed papers, their methodologies, key findings, and relevance to our project, following the literature table format recommended by Michigan State University Libraries [23].

Table 2.1: Literature review summary table.

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Werner et al. (2022) [1]	DeFi protocol systematization	SoK survey of 12+ DeFi protocol categories	Lending platforms are uniformly pool-based and over-collateralized; no institutional hierarchy exists	Identifies the core gap our four-tier architecture addresses

Continued on next page

Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Bastankhah et al. (2023) [2]	Adaptive DeFi lending	Dual fast/slow control; simulation on historical data	Dynamic rate adjustment outperforms static utilization curves	Validates algorithmic lending parameter optimization
Palaiokrassas et al. (2023) [3]	ML fraud detection in DeFi	XGBoost, NN classifiers on 54M+ multi-chain transactions	DeFi behavioral features yield F1: 0.76–0.85 vs. 0.08 with transaction features alone	Informs our feature engineering for Random Forest fraud model
Adom et al. (2022) [4]	XAI in loan approval	LIME vs. SHAP comparison on lending datasets	SHAP provides deeper, more consistent attributions; LIME is faster but less stable	Justifies SHAP as our primary explainability method
Tan (2023) [5]	CBDC and financial inclusion	IMF economic model for developing nations	Two-tier CBDC distribution banks large unbanked populations	Directly parallels our four-tier institutional hierarchy
Liu et al. (2008) [6]	Anomaly detection	Isolation Forest algorithm on synthetic and real datasets	Isolation via recursive partitioning; shorter paths for anomalies	Adopted as secondary unsupervised detection for wallet behavior

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Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Atzei et al. (2017) [7]	Smart contract security	SoK of Ethereum attack vectors	Catalogs reentrancy, overflow, access control vulnerabilities	Directly informs our security primitives and planned formal verification

2.4 Summary of Key Findings

The literature review yields the following consolidated findings that directly inform our design:

- **DeFi lending is structurally flat.** Existing protocols (Aave, Compound, MakerDAO) use pool-based or over-collateralized models. No protocol models institutional hierarchy analogous to development finance. CBDC research [5] confirms that tiered distribution is viable and supports financial inclusion.
- **ML improves fraud detection in blockchain contexts.** DeFi-specific behavioral features substantially outperform transactional features alone (F1: 0.76–0.85 vs. 0.08 [3]). Isolation Forest [6] enables unsupervised detection where labeled data is scarce.
- **SHAP satisfies regulatory explainability.** SHAP provides more consistent feature attributions than LIME for loan approval systems [4], meeting prudential requirements for transparent automated decision-making.
- **Smart contract security requires layered defense.** Reentrancy, overflow, and access control vulnerabilities are well-documented [7]. OpenZeppelin primitives and formal verification tools are recommended for production deployments.
- **Blockchain enables financial inclusion.** Systematic reviews confirm that blockchain and CBDC infrastructure can reach the 1.4 billion unbanked adults globally [14] through tiered distribution models [5].

These findings justify our four-tier architecture, AI/ML integration (Random Forest, Isolation Forest, SHAP), and governance design.

Chapter 3

System Architecture and Design

3.1 High-Level Architecture

The system employs a **three-layer decentralized application architecture**:

PRESENTATION LAYER

React 18 · TypeScript · Material Design 3 · Wagmi · Viem

Modules: Dashboard, Deposit, Loan, Admin, Risk AI, QR

↓

SMART CONTRACT LAYER (EVM: Ethereum / Polygon)

World Bank Reserve Contract · National Bank Contract · Local Bank Contract

Operations: Reserve Mgmt, Hierarchical Lending, Loan Lifecycle, Access Control, Emergency Controls

↓

OFF-CHAIN SERVICES LAYER

PostgreSQL (Relational DB) · **FastAPI** (REST API)

AI/ML: Random Forest (Fraud) · Isolation Forest (Anomaly) · SHAP (Explainability)

Event Listener · Cache (Redis)

Figure 3.1 shows the component interactions across these three layers.

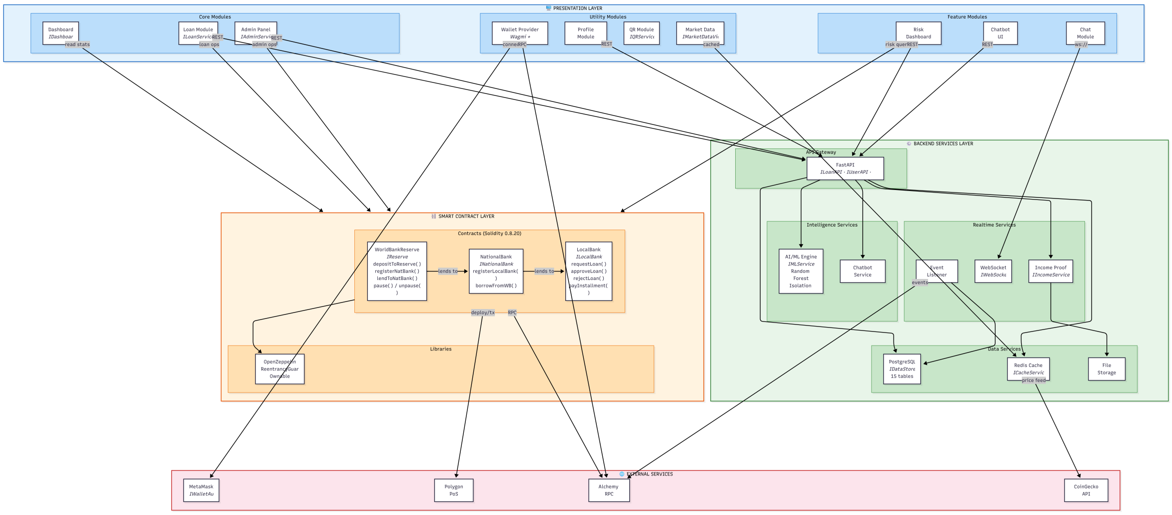


Figure 3.1: Component diagram showing interactions between the presentation layer, smart contract layer, off-chain backend services, and external systems.

3.2 Blockchain Platform Selection

Table 3.1: Blockchain platform selection criteria and justification.

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling
Network	Polygon Mumbai / Ethereum Sepolia (testnets)	Zero-cost deployment; production-equivalent behavior; free faucet access
Consensus	Proof-of-Stake (PoS) via Polygon validators	Energy-efficient; sub-2-second block finality; decentralized validator set
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem

3.2.1 Transaction Verification and Consensus

- **Polygon PoS:** Transactions are validated by a decentralized set of PoS validators. Block finality is achieved within approximately 2 seconds. Checkpoints are periodically committed to Ethereum mainnet for additional security.

- **On prototype testnets:** Mumbai and Sepolia employ analogous consensus models with no financial cost, enabling iterative development and testing.

3.3 Data Model and Database Design

The relational database schema comprises 15 normalized entities in Third Normal Form (3NF). Figure 3.2 presents the core system graph showing entity relationships, Figure 3.3 presents the full Entity-Relationship Diagram (ERD), and Figure 3.4 presents the Enhanced Entity-Relationship (EER) diagram.

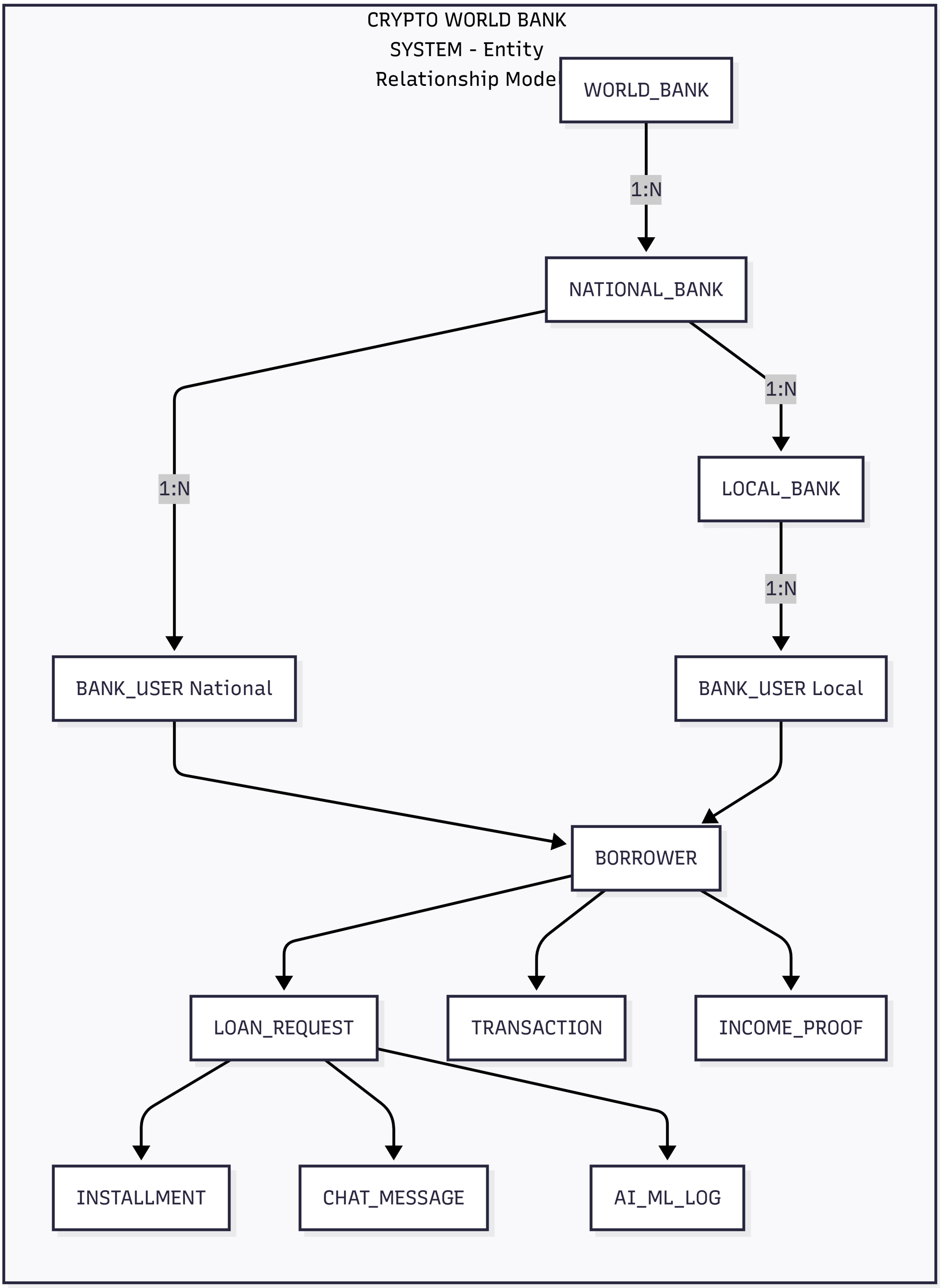


Figure 3.2: Core system graph showing the Crypto World Bank entity relationship model with hierarchical banking structure, lending lifecycle, communication, and security entities.

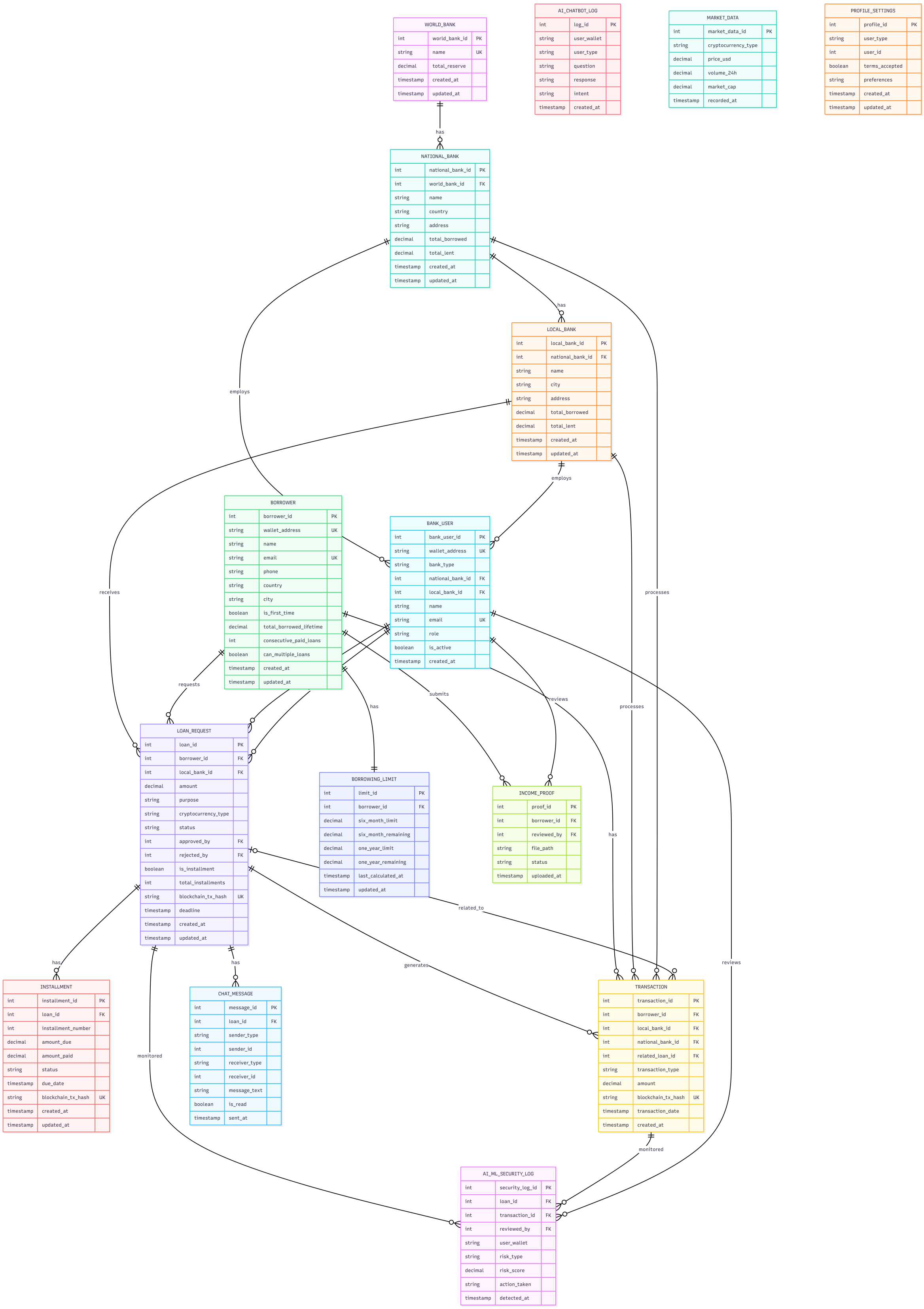


Figure 3.3: Entity-Relationship Diagram (ERD) for the Crypto World Bank database with relational symbols and entity attributes across all 15 tables.

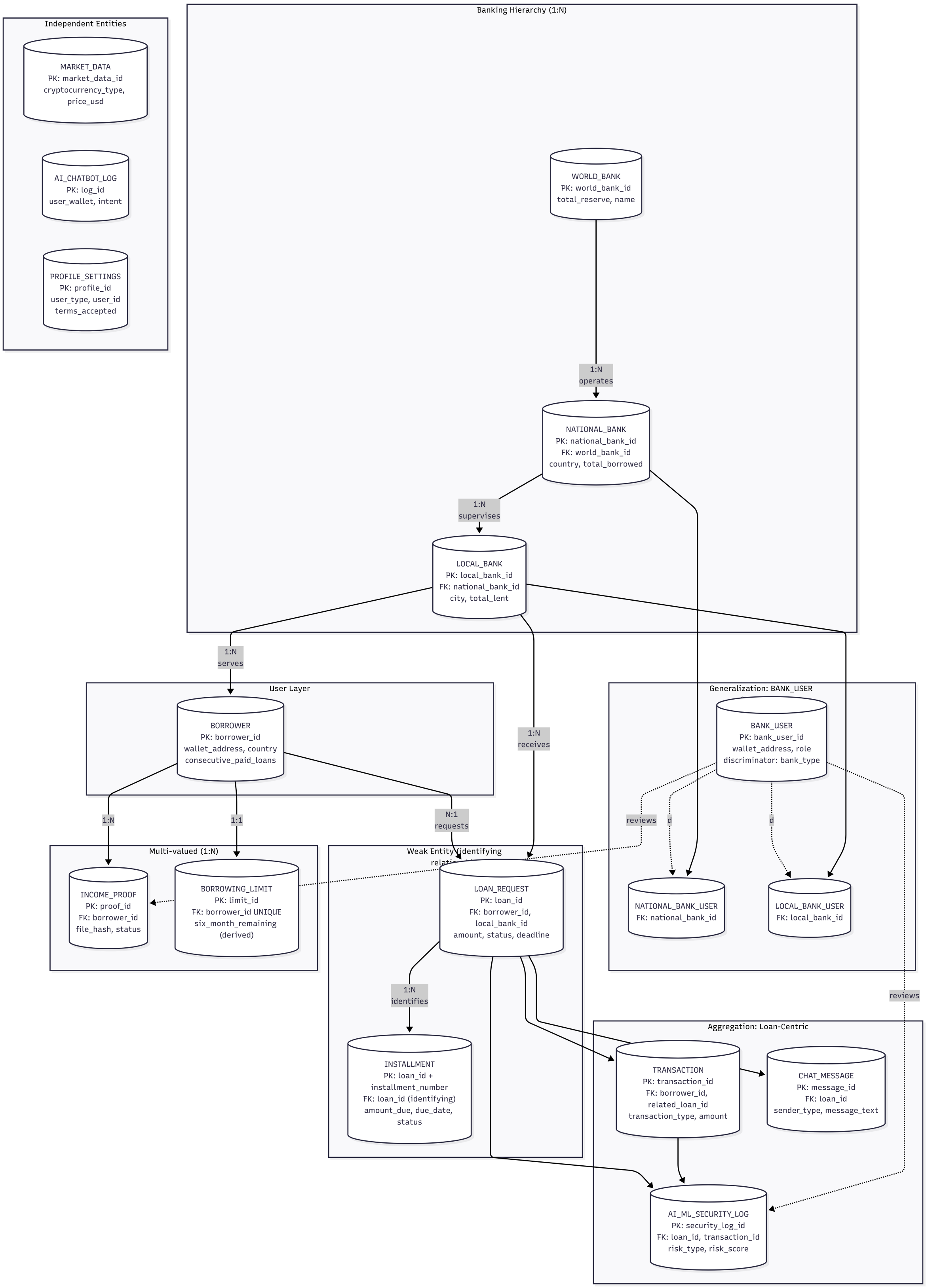


Figure 3.4: Enhanced Entity-Relationship (EER) diagram showing generalization/specialization, weak entities, aggregation, participation constraints, and cardinality for the complete data model.

3.3.1 Entity Summary

Table 3.2: Database entity summary (15 entities).

Entity	Role
WORLD_BANK	Top-level reserve holder; global lending parameters
NATIONAL_BANK	Country-level banks; borrow from World Bank
LOCAL_BANK	City-level banks; retail lending to borrowers
BANK_USER	Bank staff with role-based permissions (approve/reject loans)
BORROWER	End-users requesting and repaying loans
LOAN_REQUEST	Loan applications and full lifecycle tracking
INSTALLMENT	Installment payment records (weak entity dependent on LOAN_REQUEST)
TRANSACTION	Financial transaction records
BORROWING_LIMI	Per-borrower limits with derived 6-month/1-year windows
INCOME_PROOF	Income verification documents (multi-valued per borrower)
CHAT_MESSAGE	Borrower–bank communication
AI_CHATBOT_LOG	AI chatbot interaction records
AI_ML_SECURITY_	Security and ML monitoring events
MARKET_DATA	Cryptocurrency price feeds
PROFILE_SETTING	User profile and platform preferences

3.3.2 EER Constructs Applied

Table 3.3: EER constructs applied in the Crypto World Bank data model.

EER Construct	Application
Generalization (disjoint)	BANK_USER generalizes to NATIONAL_BANK_USER and LOCAL_BANK_USER via `bank_type` discriminator with mutually exclusive foreign keys
Weak entity	INSTALLMENT is existence-dependent on LOAN_REQUEST with partial key `installment_number`
Aggregation	LOAN_REQUEST aggregates BORROWER + LOCAL_BANK; TRANSACTION, CHAT_MESSAGE, and AI_ML_SECURITY_LOG are components
Multi-valued attribute	INCOME_PROOF modeled as a separate entity (borrower can have multiple proofs)
Derived attribute	`six_month_remaining` and `one_year_remaining` in BORROWING_LIMIT are computed from TRANSACTION history
Composite key	INSTALLMENT uses (`loan_id`, `installment_number`)
Participation constraints	BORROWER has total participation in LOAN_REQUEST; INSTALLMENT has total participation when `is_installment = TRUE`
Cardinality	1:1 (BORROWER–BORROWING_LIMIT), 1:N (hierarchy, loan–installment), N:1 (loan–borrower)

3.3.3 Normalization

The schema is normalized to Third Normal Form (3NF) with verification against Boyce-Codd Normal Form (BCNF):

- **1NF:** All attributes are atomic with no repeating groups. Multi-valued attributes like income proofs are stored in separate entities.
- **2NF:** No partial dependencies exist. INSTALLMENT's non-key attributes depend on the full composite key (`loan_id`, `installment_number`).

- **3NF:** No transitive dependencies. Derived values such as `total_borrowed` in bank entities are computed at query time rather than stored redundantly.
- **BCNF:** All determinants are candidate keys. A CHECK constraint on BANK_USER enforces disjoint specialization of the generalization hierarchy.

3.3.4 Indexing Strategy

We use B-tree indexes (the default in PostgreSQL) for efficient retrieval on frequently queried columns:

Table 3.4: Database indexing strategy.

Table	Index	Purpose
LOAN_REQUEST	`idx_status_borrower`, `idx_status_bank`, `idx_deadline`	Filter pending loans; lookup by borrower or bank
TRANSACTION	`idx_type_date`, `idx_borrower_date`, `idx_date`	Borrowing limit queries (6-month/1-year windows)
BORROWER	`idx_wallet`, `idx_location`	Wallet address lookup; location-based queries
CHAT_MESSAGE	`idx_loan_sent`, `idx_sender`, `idx_receiver`	Chat history retrieval; unread message counts

B-tree indexes are chosen for their efficiency with range queries (e.g., transactions within the last 6 months); hash indexes would be suitable only for exact-match lookups.

3.3.5 Functional Dependencies

Table 3.5: Representative functional dependencies.

Relation	Functional Dependency	Notes
LOAN_REQUEST	`loan_id` → all attributes	Primary key determines row
LOAN_REQUEST	`borrower_id`, `local_bank_id` → `status`, `amount`, ...	One active request per borrower per bank
BORROWING_LI	`borrower_id` → `six_month_limit`, `one_year_limit`, ...	1:1 with BORROWER

3.3.6 Relational Integrity Constraints

Table 3.6: Relational integrity constraints.

Constraint Type	Examples
Primary Key	`world_bank_id`, `loan_id`, `borrower_id`, etc.
Foreign Key	`local_bank_id` in LOAN_REQUEST references LOCAL_BANK
UNIQUE	`wallet_address` in BORROWER; `blockchain_tx_hash` in LOAN_REQUEST
CHECK	BANK_USER: `(bank_type='national' AND national_bank_id IS NOT NULL) OR (bank_type='local' AND local_bank_id IS NOT NULL)`
NOT NULL	Core attributes: `name`, `wallet_address`, `status`

3.4 On-Chain and Off-Chain Data Partitioning

Table 3.7: Data partitioning between on-chain and off-chain storage.

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification
User profiles, income verification documents, chat messages, AI/ML inference logs	Off-chain (PostgreSQL)	Data privacy, query flexibility, storage cost optimization
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints
Cryptocurrency market data	Off-chain (cached)	High-frequency updates; external API dependency

3.5 Digital Identity System

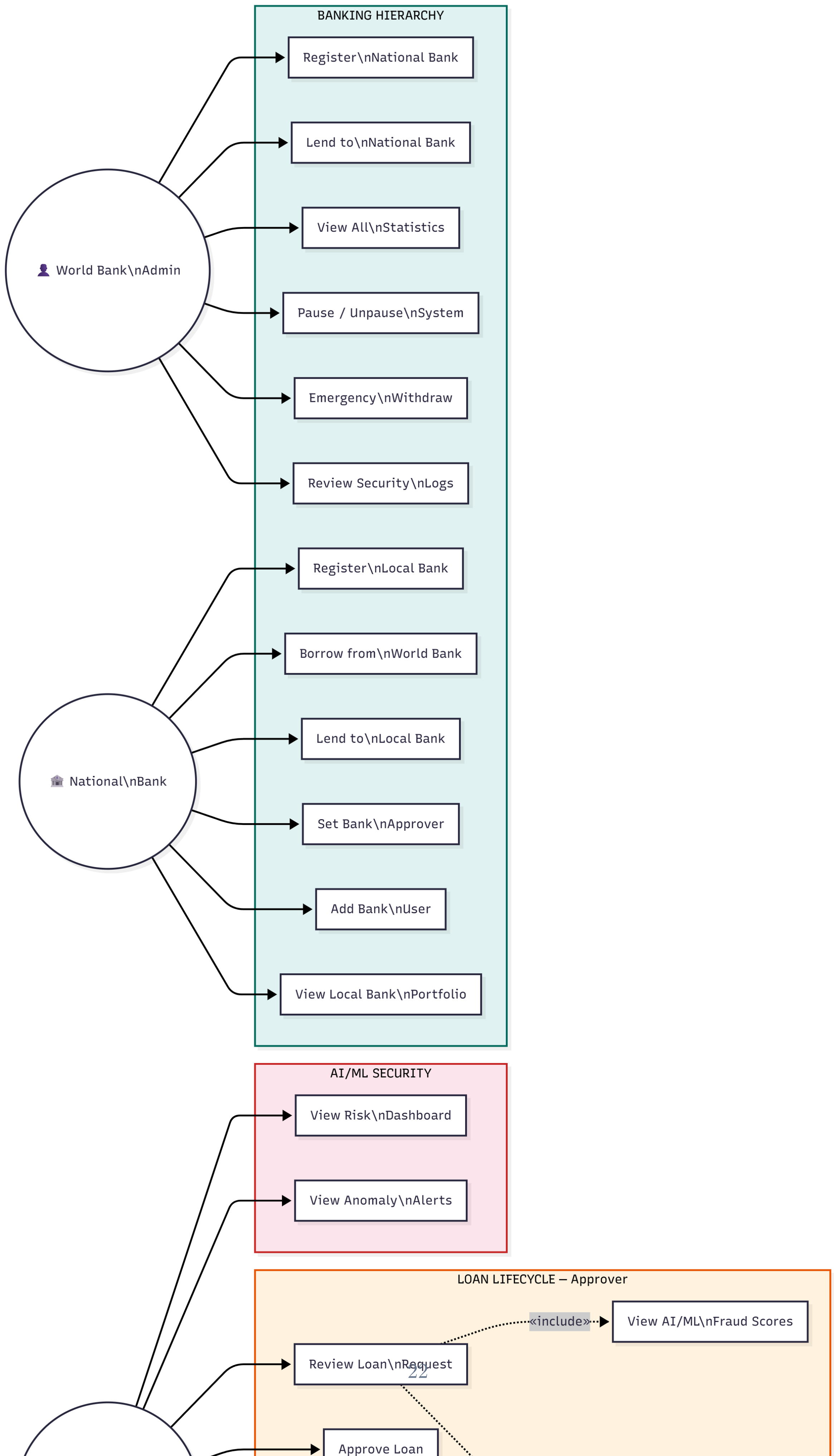
- **Wallet-based identity:** Users authenticate via Ethereum wallet signatures (MetaMask, WalletConnect). No centralized credential store is required.
- **Role binding:** Wallet addresses are mapped to hierarchical roles (Owner, National Bank, Local Bank, Approver, Borrower) within the smart contract permission system.

3.6 System Modeling

This section presents the system analysis diagrams that model the platform's structure, behavior, and data flow.

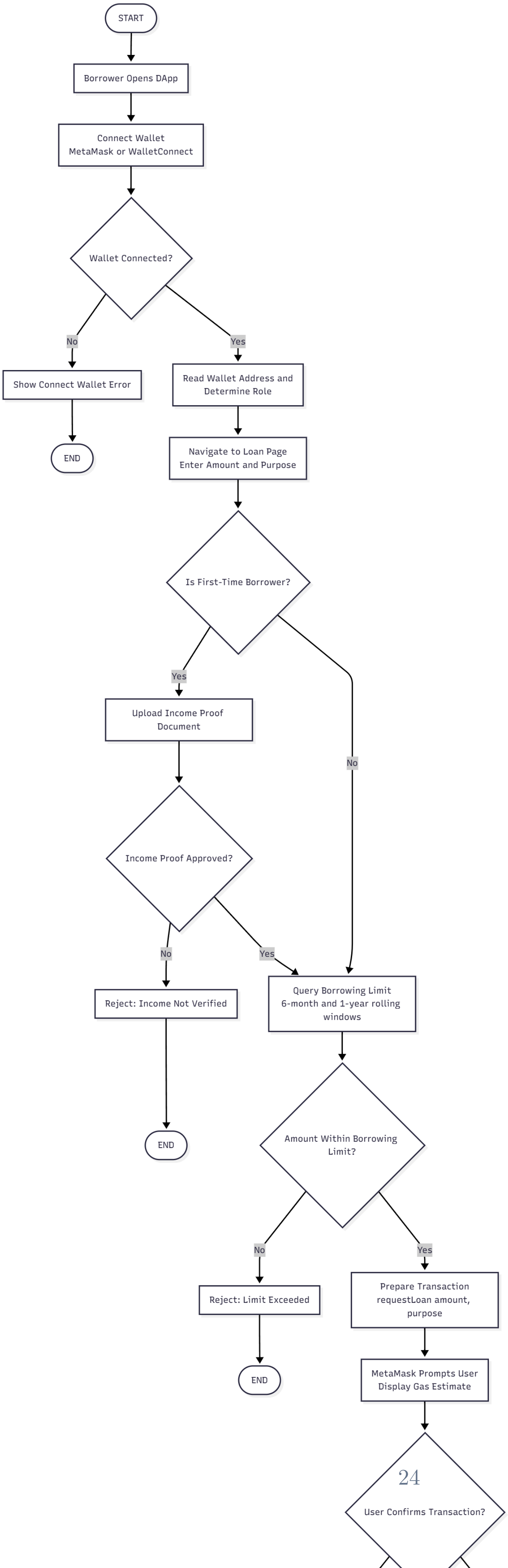
3.6.1 Use Case Diagram

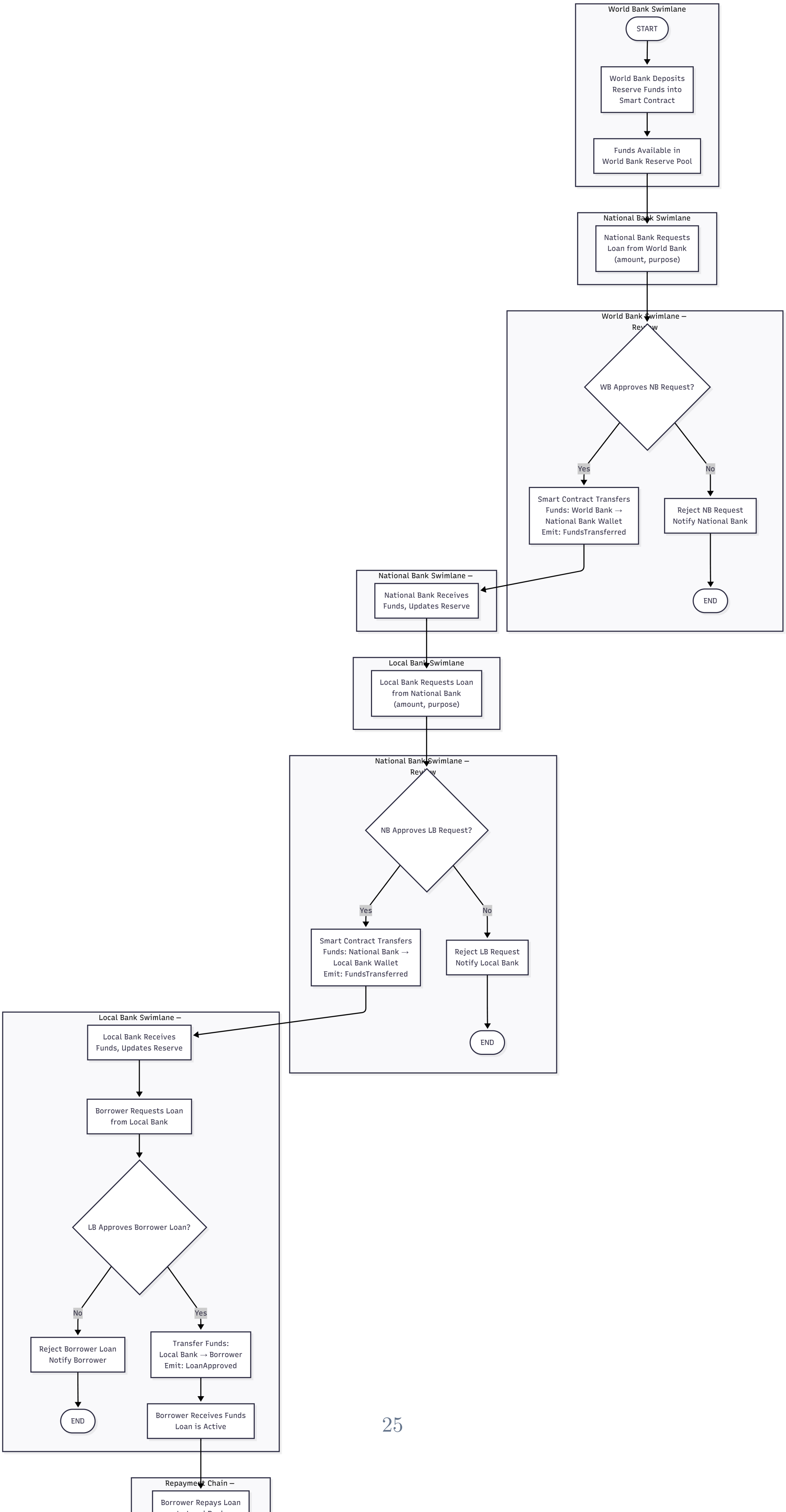
The system involves four primary actors—Borrower, Bank Approver, World Bank Admin, and National Bank—across 29 use cases. Figure 3.5 shows the complete use case diagram.

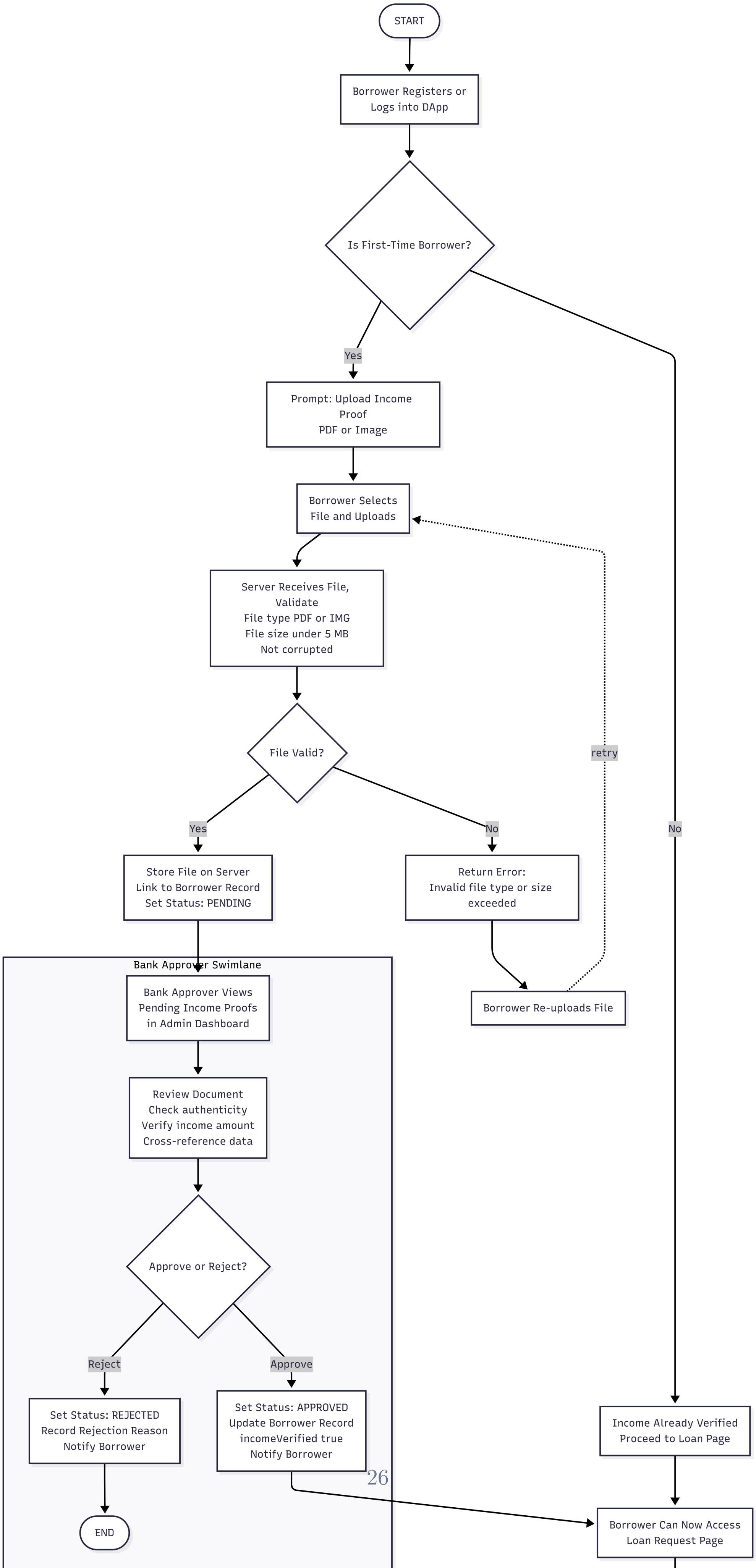

BANKING HIERARCHY
World Bank\nAdmin
Register\nNational Bank
Lend to\nNational Bank
View All\nStatistics
Pause / Unpause\nSystem
Emergency\nWithdraw
Review Security\nLogs
National\nBank
Register\nLocal Bank
Borrow from\nWorld Bank
Lend to\nLocal Bank
Set Bank\nApprover
Add Bank\nUser
View Local Bank\nPortfolio
AI/ML SECURITY
View Risk\nDashboard
View Anomaly\nAlerts
LOAN LIFECYCLE — Approver
«include»
View AI/ML\nFraud Scores
Review Loan\nRequest
Approve Loan

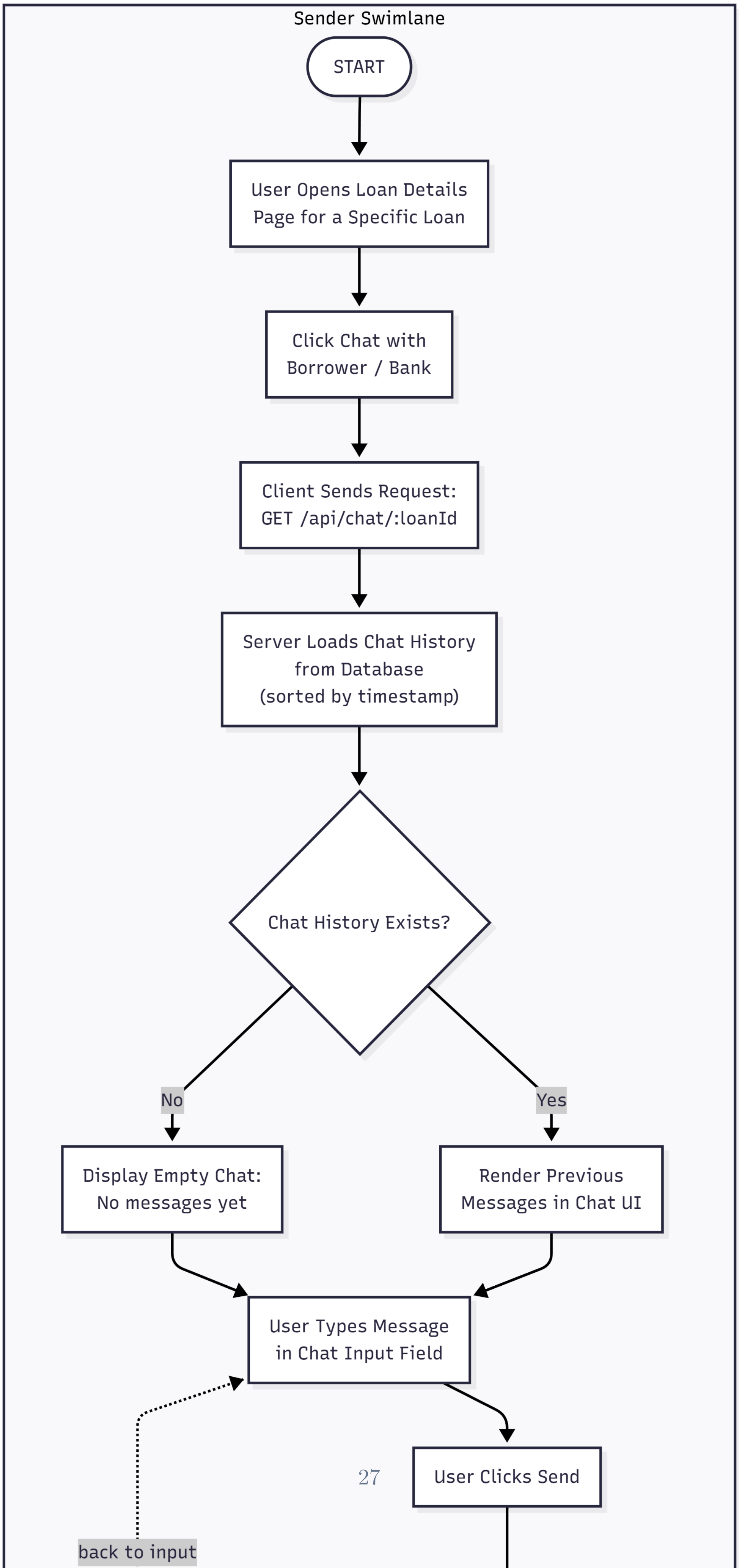
3.6.2 Activity Diagrams

Figures 3.6–3.12 present the activity diagrams for the platform's seven core operational flows.


START
Borrower Opens DApp
Connect Wallet
MetaMask or WalletConnect
Wallet Connected?
No
Yes
Show Connect Wallet Error
END
Read Wallet Address and Determine Role
Navigate to Loan Page
Enter Amount and Purpose
Is First-Time Borrower?
Yes
No
Upload Income Proof Document
Income Proof Approved?
No
Yes
Reject: Income Not Verified
END
Query Borrowing Limit
6-month and 1-year rolling windows
Amount Within Borrowing Limit?
No
Yes
Reject: Limit Exceeded
END
Prepare Transaction
requestLoan amount, purpose
MetaMask Prompts User
Display Gas Estimate
User Confirms Transaction?


World Bank Swimlane
START
World Bank Deposits
Reserve Funds into
Smart Contract
Funds Available in
World Bank Reserve Pool
National Bank Swimlane
National Bank Requests
Loan from World Bank
(amount, purpose)
World Bank Swimlane –
WB Approves NB Request?
Yes
No
Smart Contract Transfers
Funds: World Bank →
National Bank Wallet
Emit: FundsTransferred
Reject NB Request
Notify National Bank
END
National Bank Swimlane –
National Bank Receives
Funds, Updates Reserve
Local Bank Swimlane
Local Bank Requests Loan
from National Bank
(amount, purpose)
National Bank Swimlane –
NB Approves LB Request?
Yes
No
Smart Contract Transfers
Funds: National Bank →
Local Bank Wallet
Emit: FundsTransferred
Reject LB Request
Notify Local Bank
END
Local Bank Swimlane –
Local Bank Receives
Funds, Updates Reserve
Borrower Requests Loan
from Local Bank
LB Approves Borrower Loan?
No
Yes
Reject Borrower Loan
Notify Borrower
Transfer Funds:
Local Bank → Borrower
Emit: LoanApproved
END
Borrower Receives Funds
Loan is Active
Repayment Chain –
Borrower Repays Loan


START
Borrower Registers or Logs into DApp
Is First-Time Borrower?
Yes
No
Prompt: Upload Income Proof PDF or Image
Borrower Selects File and Uploads
Server Receives File, Validate File type PDF or IMG File size under 5 MB Not corrupted
File Valid?
Yes
No
retry
Store File on Server Link to Borrower Record Set Status: PENDING
Return Error: Invalid file type or size exceeded
Borrower Re-uploads File
Bank Approver Swimlane
Bank Approver Views Pending Income Proofs in Admin Dashboard
Review Document Check authenticity Verify income amount Cross-reference data
Approve or Reject?
Reject
Approve
Set Status: REJECTED Record Rejection Reason Notify Borrower
Set Status: APPROVED Update Borrower Record incomeVerified true Notify Borrower
END
Income Already Verified Proceed to Loan Page
Borrower Can Now Access Loan Request Page


Sender Swimlane
START
User Opens Loan Details Page for a Specific Loan
Click Chat with Borrower / Bank
Client Sends Request: GET /api/chat/:loanId
Server Loads Chat History from Database (sorted by timestamp)
Chat History Exists?
No
Yes
Display Empty Chat: No messages yet
Render Previous Messages in Chat UI
User Types Message in Chat Input Field
User Clicks Send
back to input

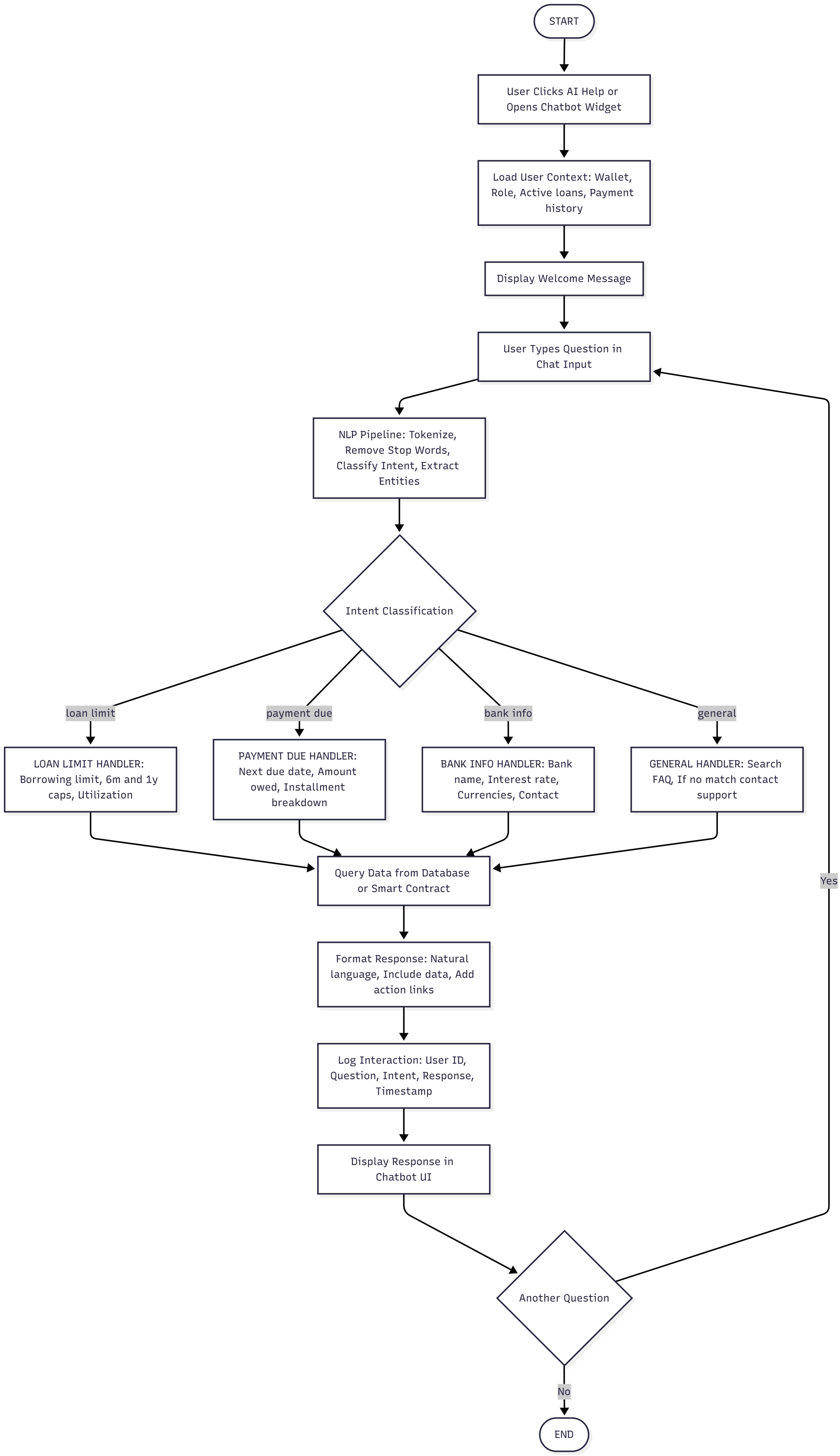
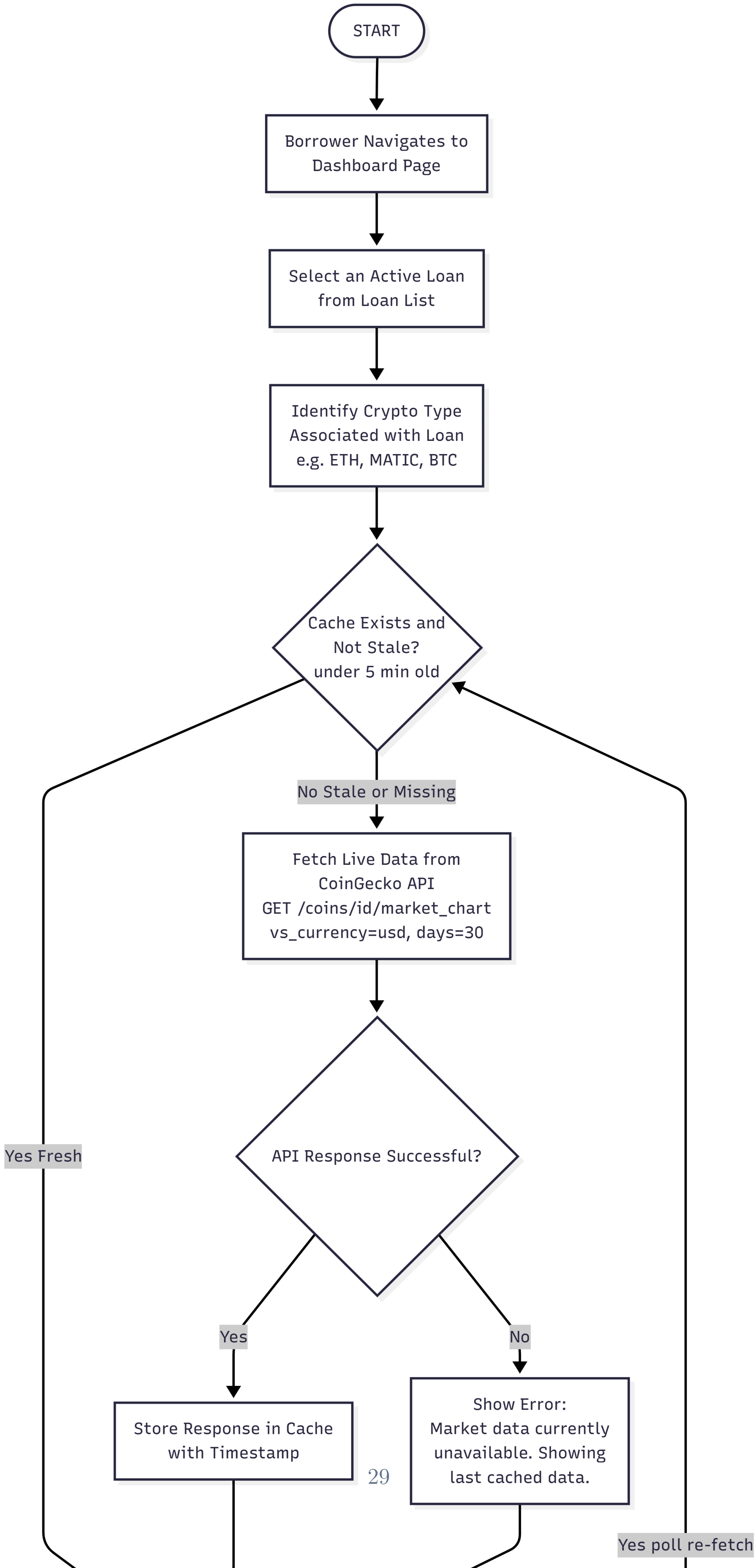


Figure 3.10: Activity diagram: AI chatbot interaction flow for automated borrower assistance and FAQ handling.


START
Borrower Navigates to Dashboard Page
Select an Active Loan from Loan List
Identify Crypto Type Associated with Loan e.g. ETH, MATIC, BTC
Cache Exists and Not Stale? under 5 min old
No Stale or Missing
Fetch Live Data from CoinGecko API GET /coins/id/market_chart vs_currency=usd, days=30
Yes Fresh
API Response Successful?
Yes
No
Store Response in Cache with Timestamp
Show Error: Market data currently unavailable. Showing last cached data.
Yes poll re-fetch

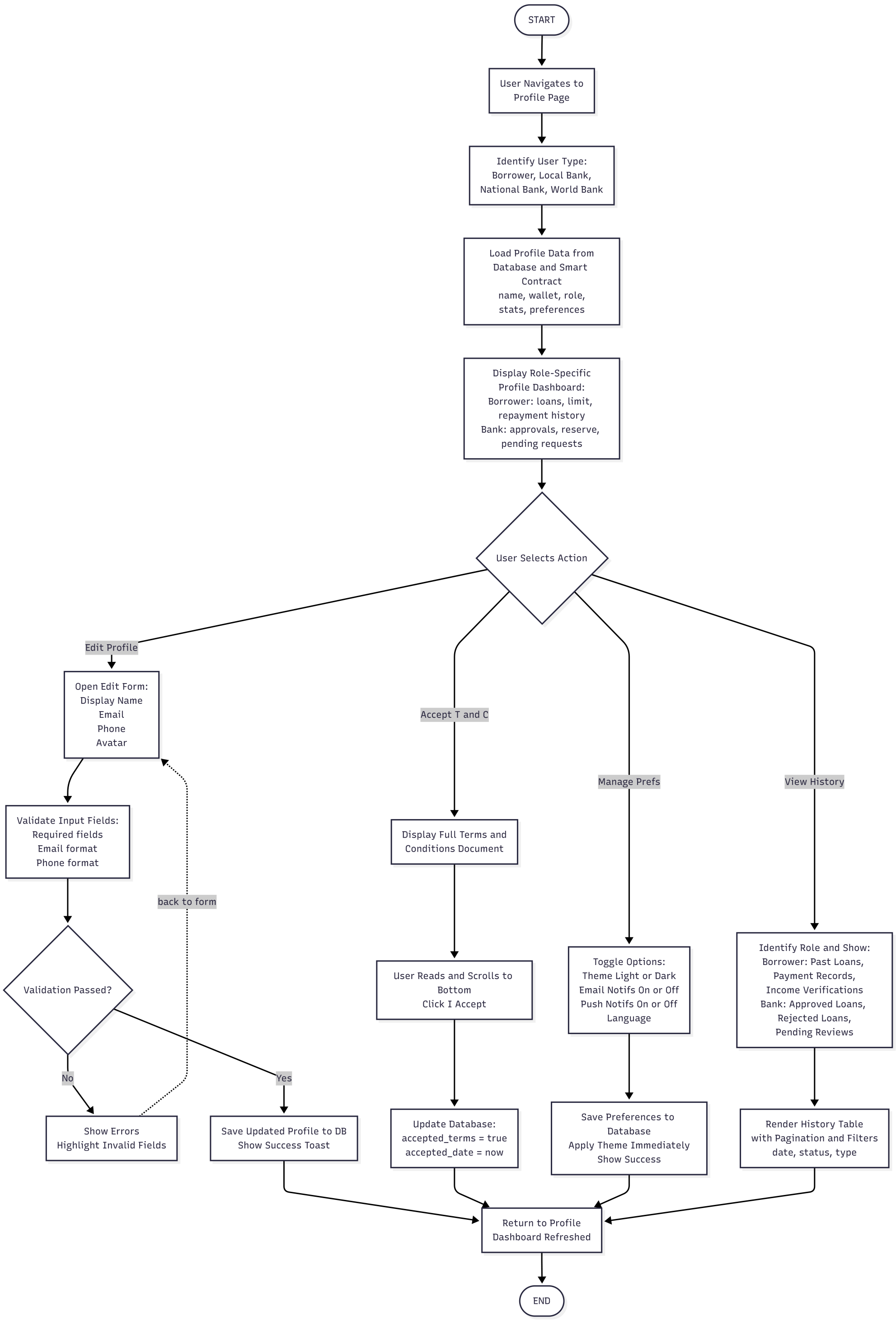


Figure 3.12: Activity diagram: Profile management flow for user settings, terms acceptance, and account configuration.

3.6.3 Data Flow Diagrams

Figures 3.13–3.15 present the data flow diagrams at the context level (Level 0) and decomposed level (Level 1).

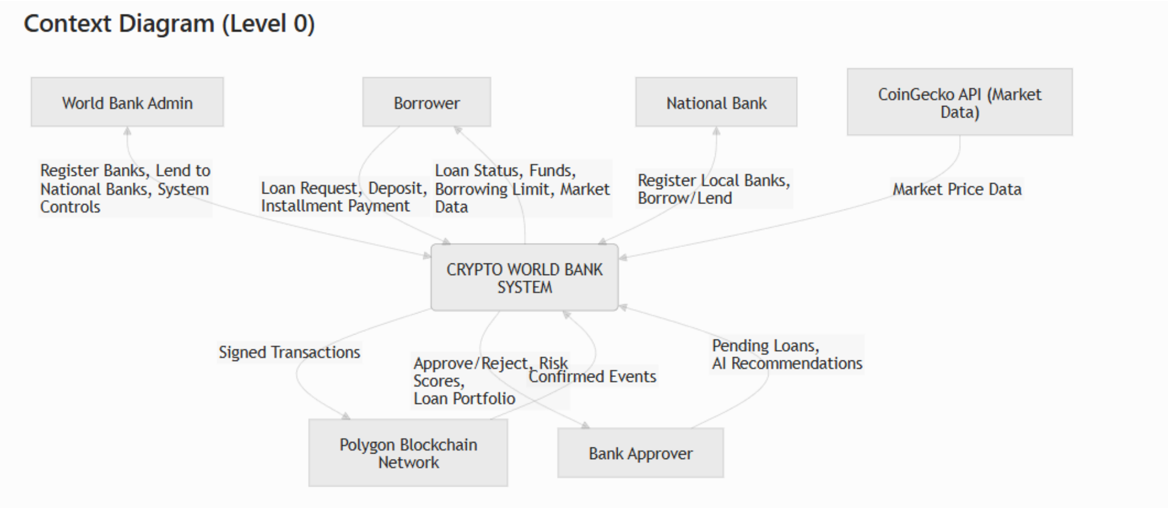


Figure 3.13: Data flow diagram — Context diagram (Level 0) showing external entities and data flows into and out of the Crypto World Bank system.

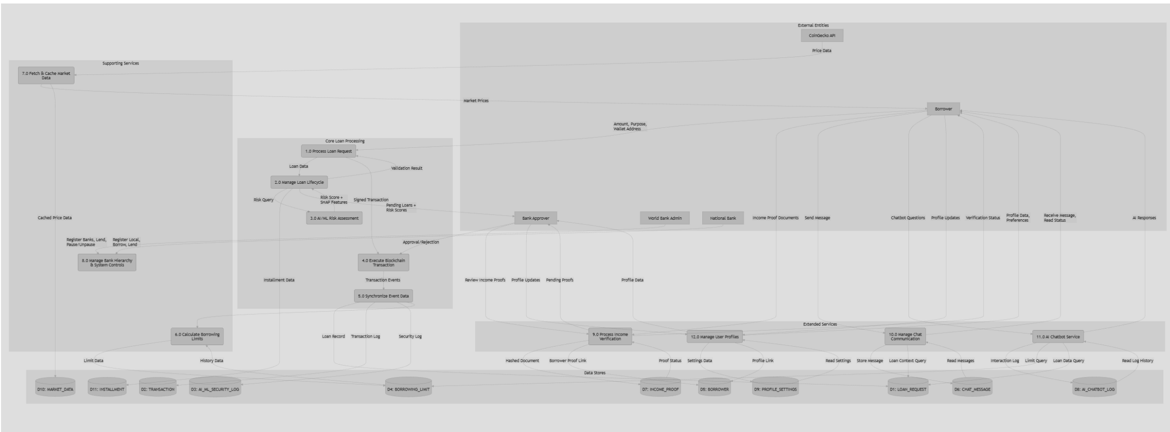


Figure 3.14: Data flow diagram — Level 1 (Part 1) showing decomposed processes, data stores, and information movement across system components.

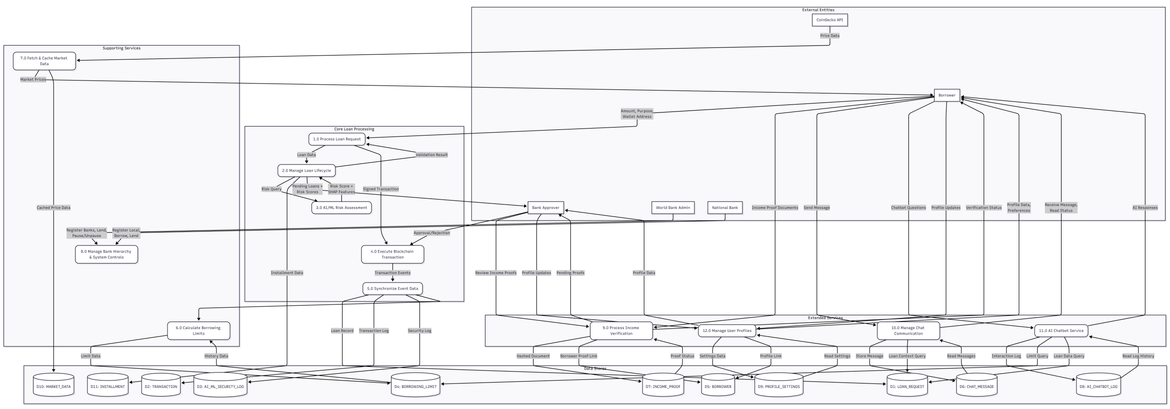


Figure 3.15: Data flow diagram — Level 1 (Part 2) showing additional processes for AI/ML analytics, market data, and profile management.

3.6.4 Sequence Diagrams

Figures 3.16–3.24 present the sequence diagrams for the platform's key interaction flows.

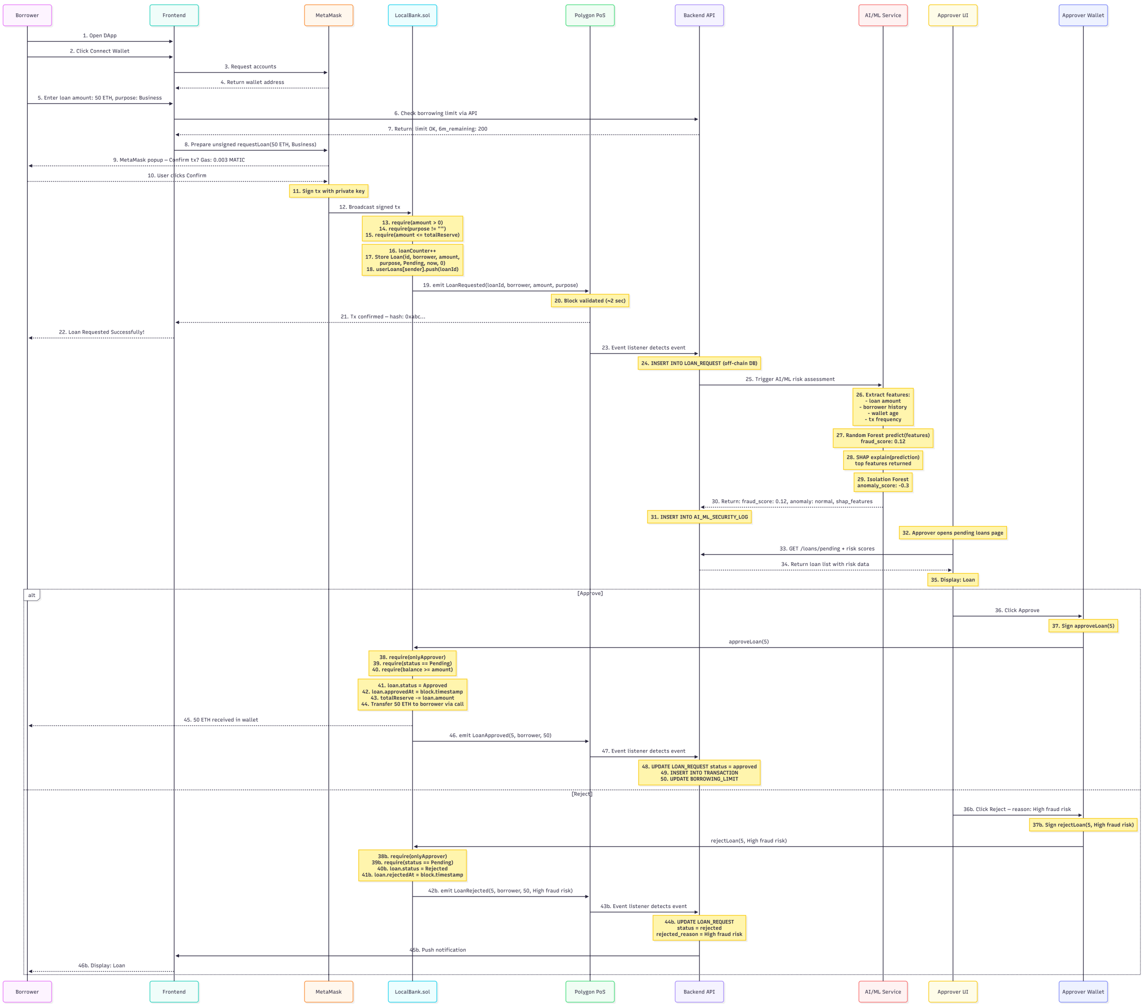


Figure 3.16: Sequence diagram: Loan request, AI risk check, and approval decision across 9 participants.

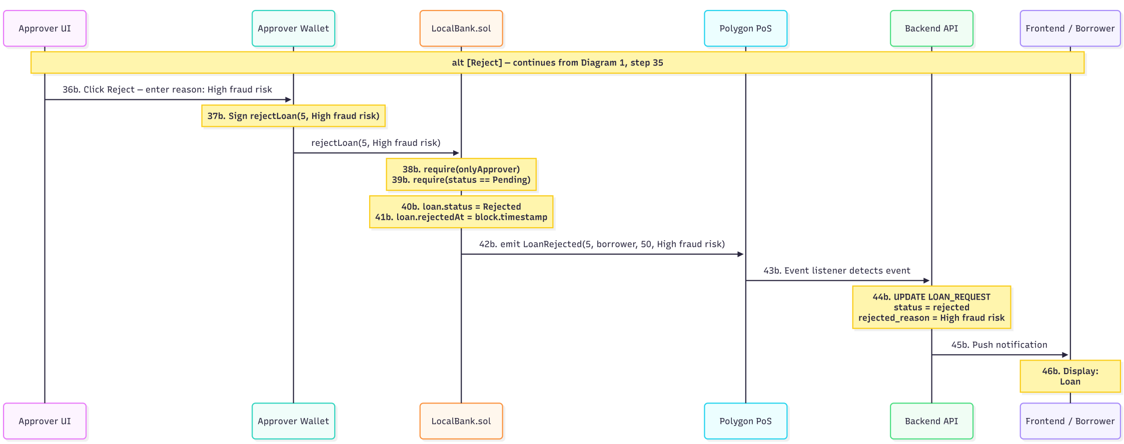


Figure 3.17: Sequence diagram: Reject path (alternative flow) for loan request denial.

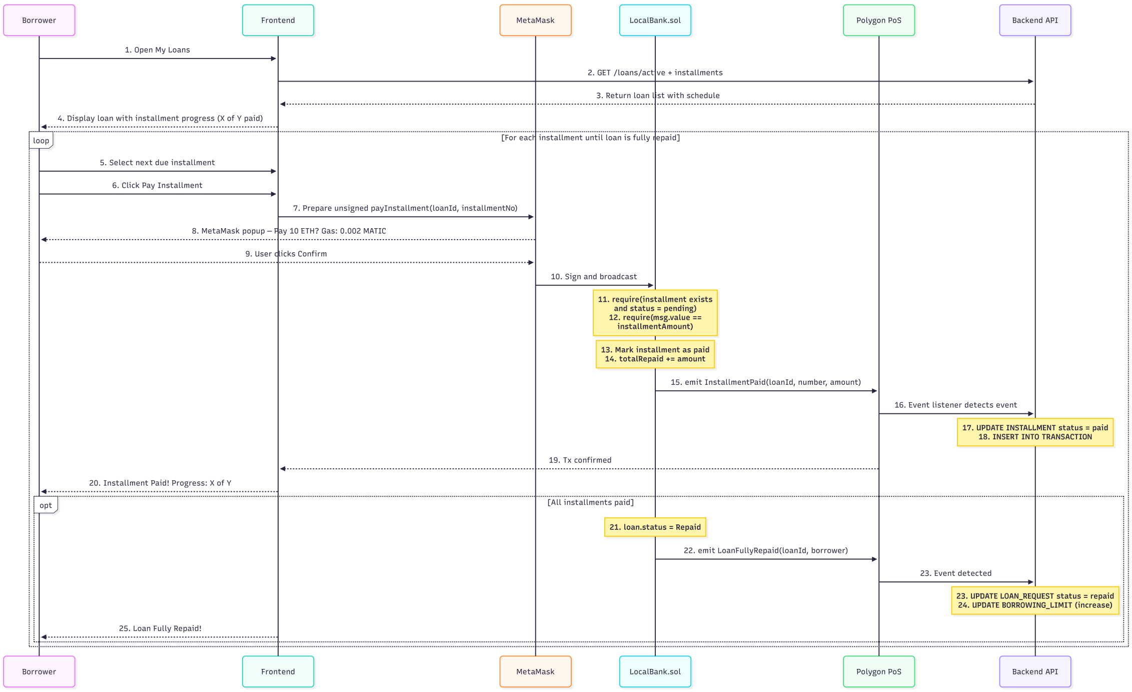


Figure 3.18: Sequence diagram: Installment payment loop showing recurring payment processing and balance updates.

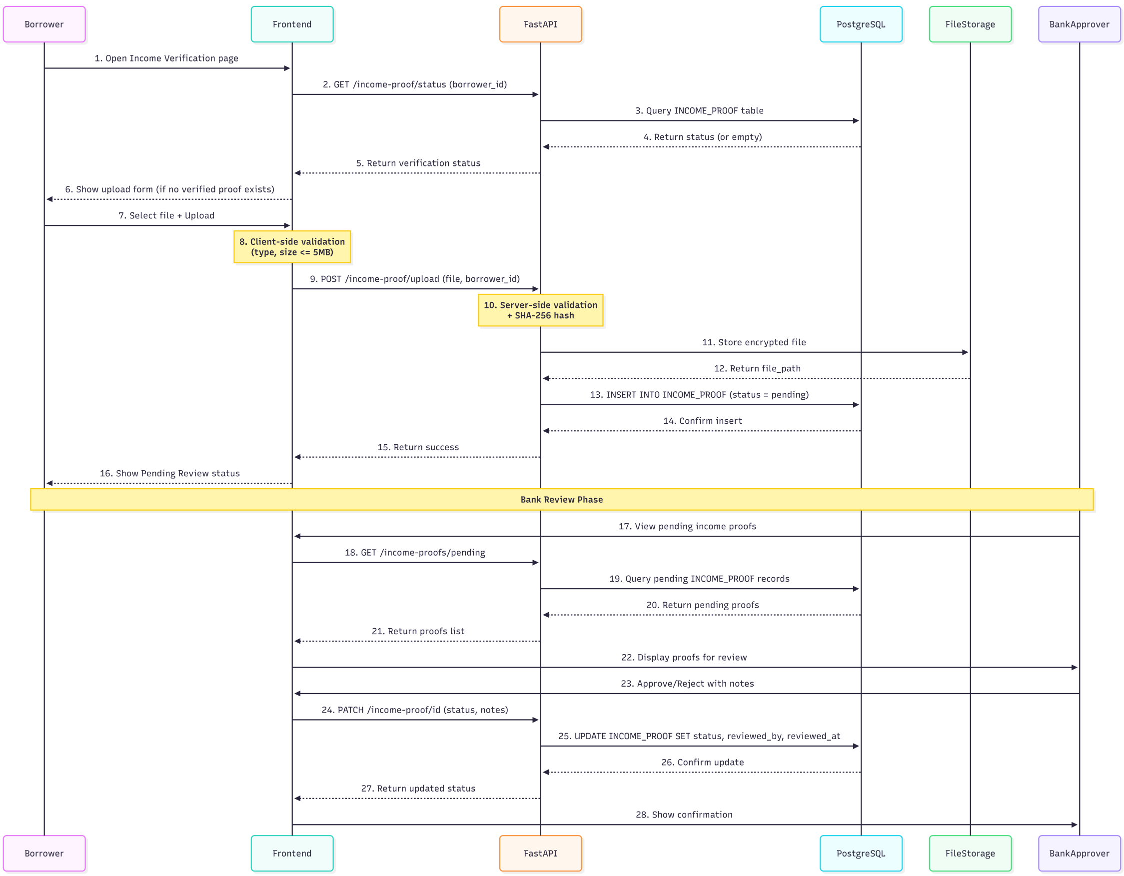


Figure 3.19: Sequence diagram: Income verification process between borrower, frontend, backend API, and database.

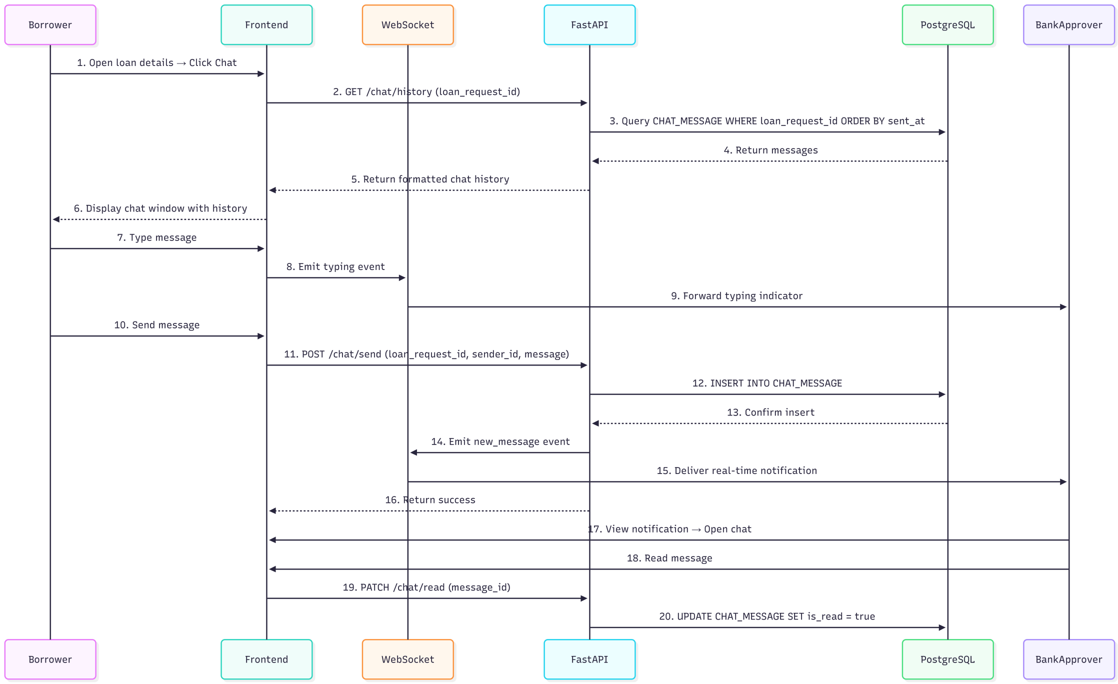


Figure 3.20: Sequence diagram: Chat system message exchange between borrower and bank representative.

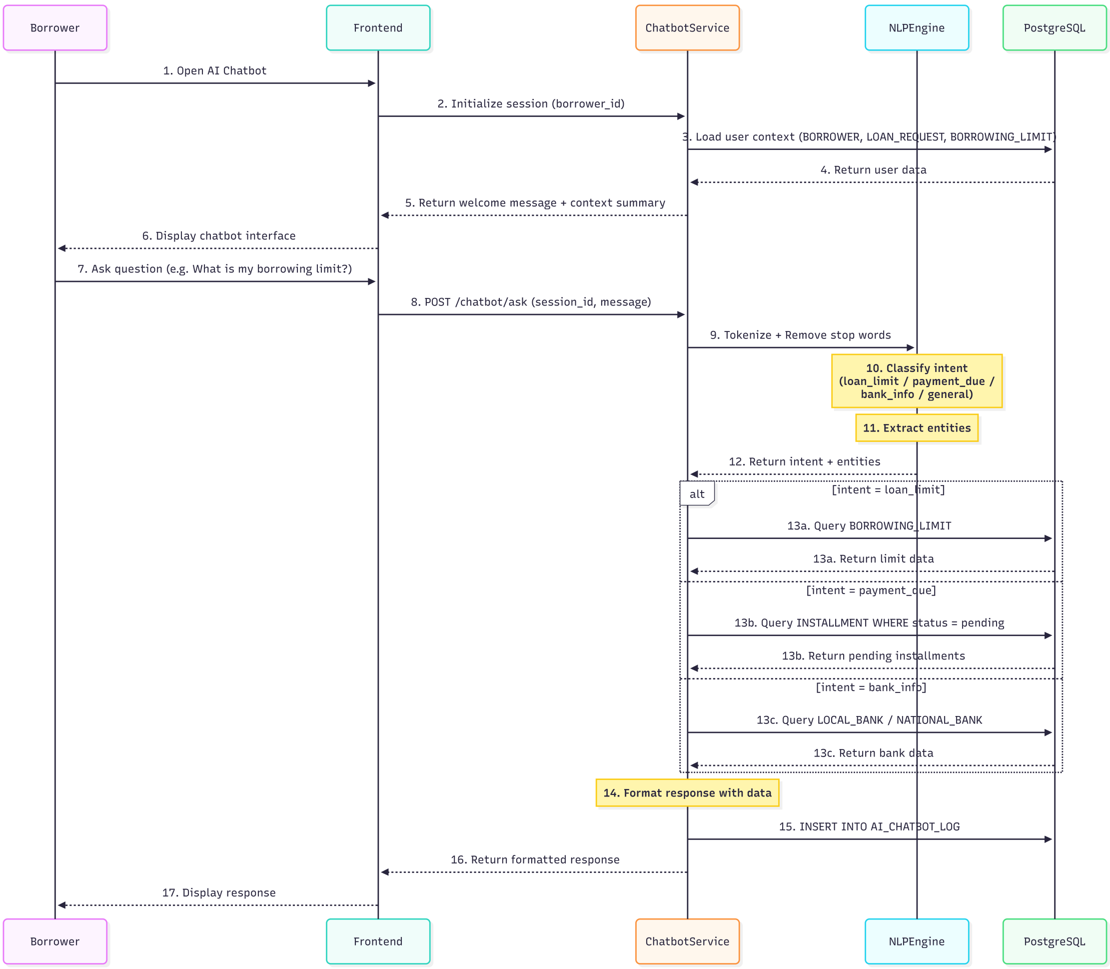


Figure 3.21: Sequence diagram: AI chatbot interaction for automated borrower support queries.

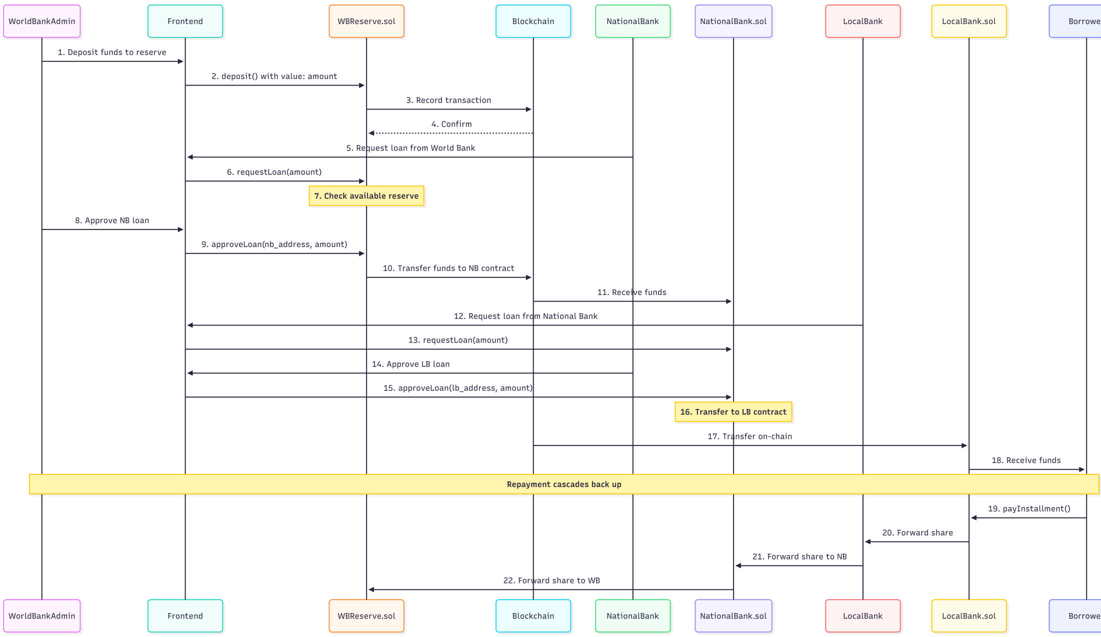


Figure 3.22: Sequence diagram: Hierarchical banking flow showing capital distribution from World Bank through National and Local Banks to Borrowers.

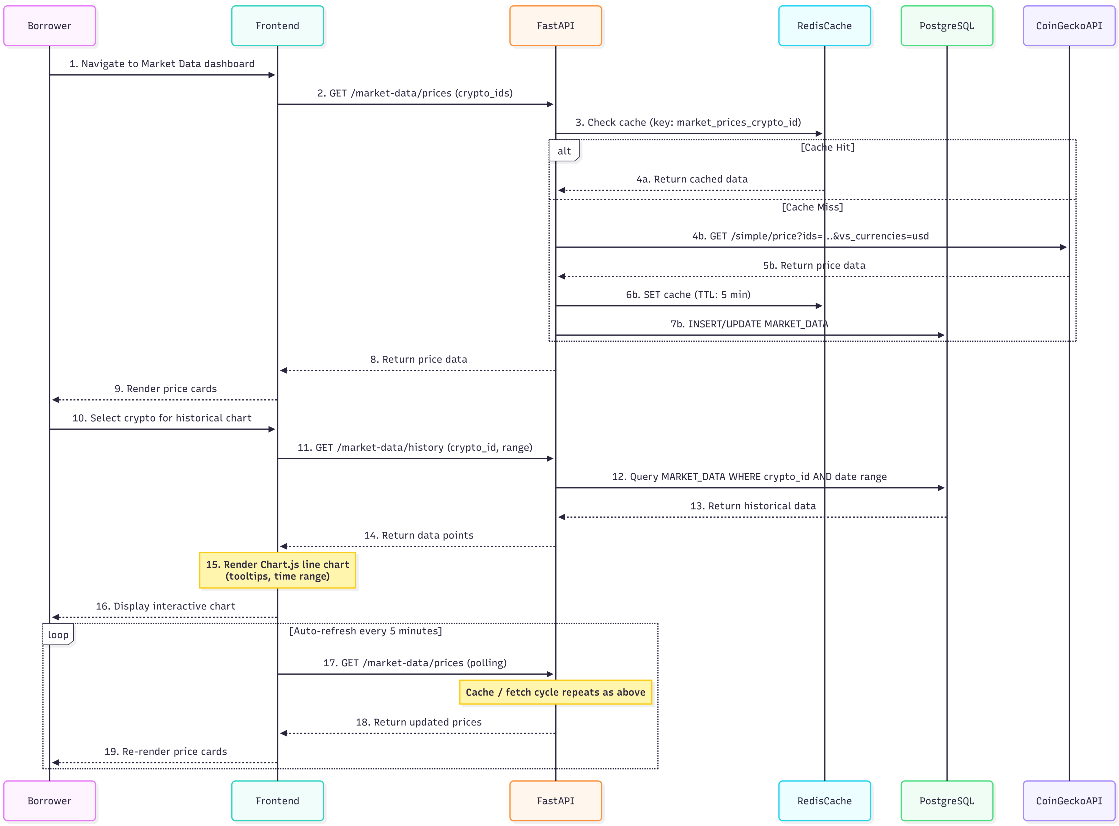


Figure 3.23: Sequence diagram: Market data retrieval from external cryptocurrency price APIs.

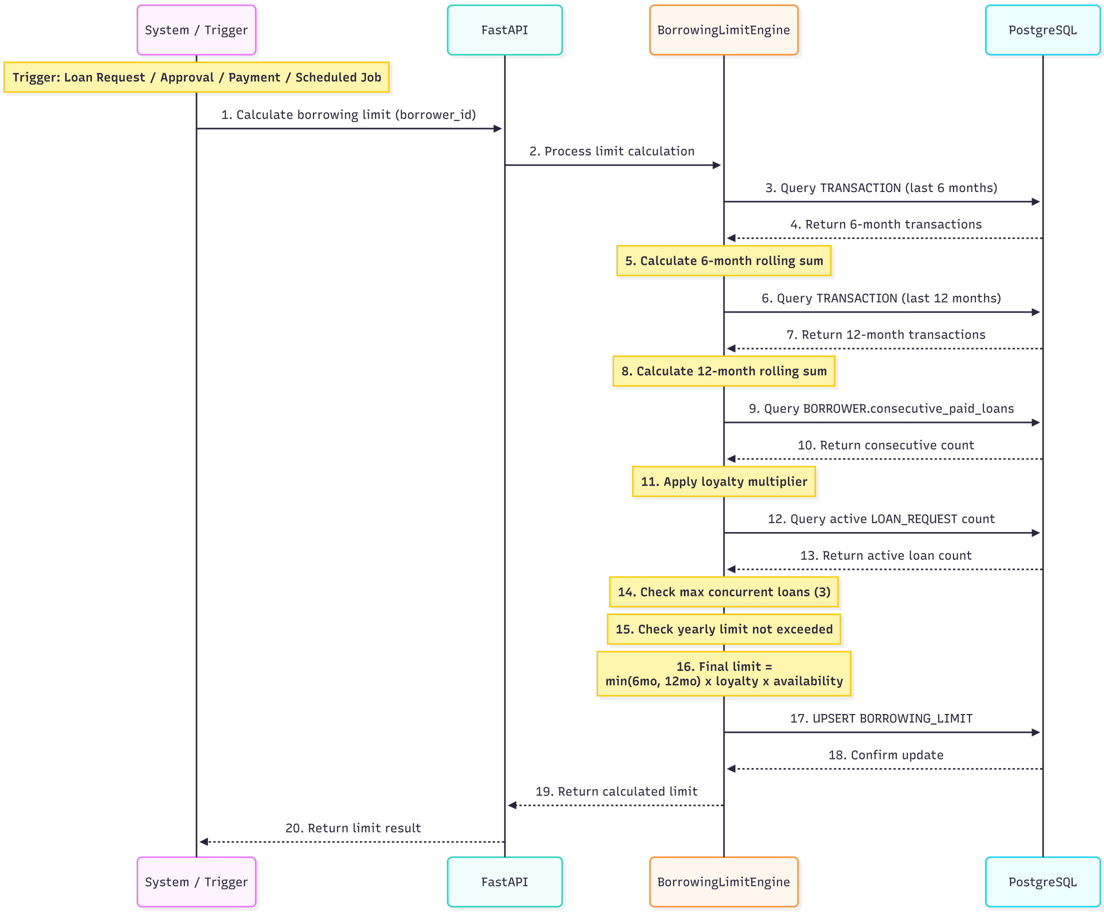


Figure 3.24: Sequence diagram: Borrowing limit calculation using rolling 6-month and 1-year transaction windows.

3.6.5 Four-Tier Capital Flow

Figure 3.25 illustrates the hierarchical capital flow and cascading repayment structure.

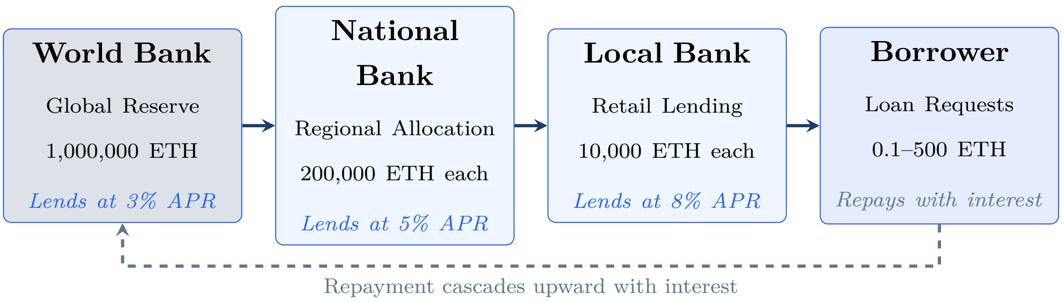


Figure 3.25: Four-tier hierarchical capital flow with cascading repayment.

3.7 Auxiliary Dual-Currency Facility

The Crypto World Bank's cryptocurrency exchange capability is designed as an **auxiliary facility** integrated into existing banking infrastructure worldwide. Rather than operating as a standalone exchange, the platform extends the services of participating banks by enabling a dual-currency feature—allowing eligible bank customers to transact in both fiat and cryptocurrency within their existing banking relationship.

- **Integration model:** The dual-currency facility is offered by all participating banks as an add-on service to their existing product portfolio. No new banking license or standalone entity is required.
- **Eligibility determination:** Eligibility for dual-currency services is determined by the bank officers who manage lending operations, based on existing KYC/AML compliance, account standing, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions, cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

3.8 Governance Framework

3.8.1 Network Membership Governance

Table 3.8: Network membership governance.

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Borrower; enforced by on-chain role-check modifiers
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations

3.8.2 Business Network Governance

Table 3.9: Business network governance.

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters codified in smart contract constants
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target uptime with multi-region deployment
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests

3.8.3 Technology Infrastructure Governance

Table 3.10: Technology infrastructure governance.

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralized delivery); blockchain layer (fully decentralized); backend API (centralized, horizontally scalable)
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimization
On-chain / off-chain data services	Clearly partitioned (see Table 3.5); event listeners synchronize state between layers
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit

3.8.4 Regulatory Compliance Considerations

As the platform operates within the regulated banking domain:

- Our prototype phase operates exclusively on **public testnets** with no real monetary value.
- The architecture supports **audit log generation** for regulatory review.
- Future production deployment would engage **regulatory sandbox programs** in target jurisdictions.

3.8.5 Asset Tokenization

- **Current implementation:** Native blockchain currency (ETH/MATIC) serves as the reserve and loan denomination.
- **Planned extension:** ERC-20 stablecoin support (USDC, USDT) for USD-denominated lending operations; tokenized collateral instruments for under-collateralized lending scenarios.

Chapter 4

Methodology

This project and its planning have been structured in accordance with the Blockchain Olympiad Bangladesh AI guidelines [16] and the final-year project evaluation rubrics.

4.1 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

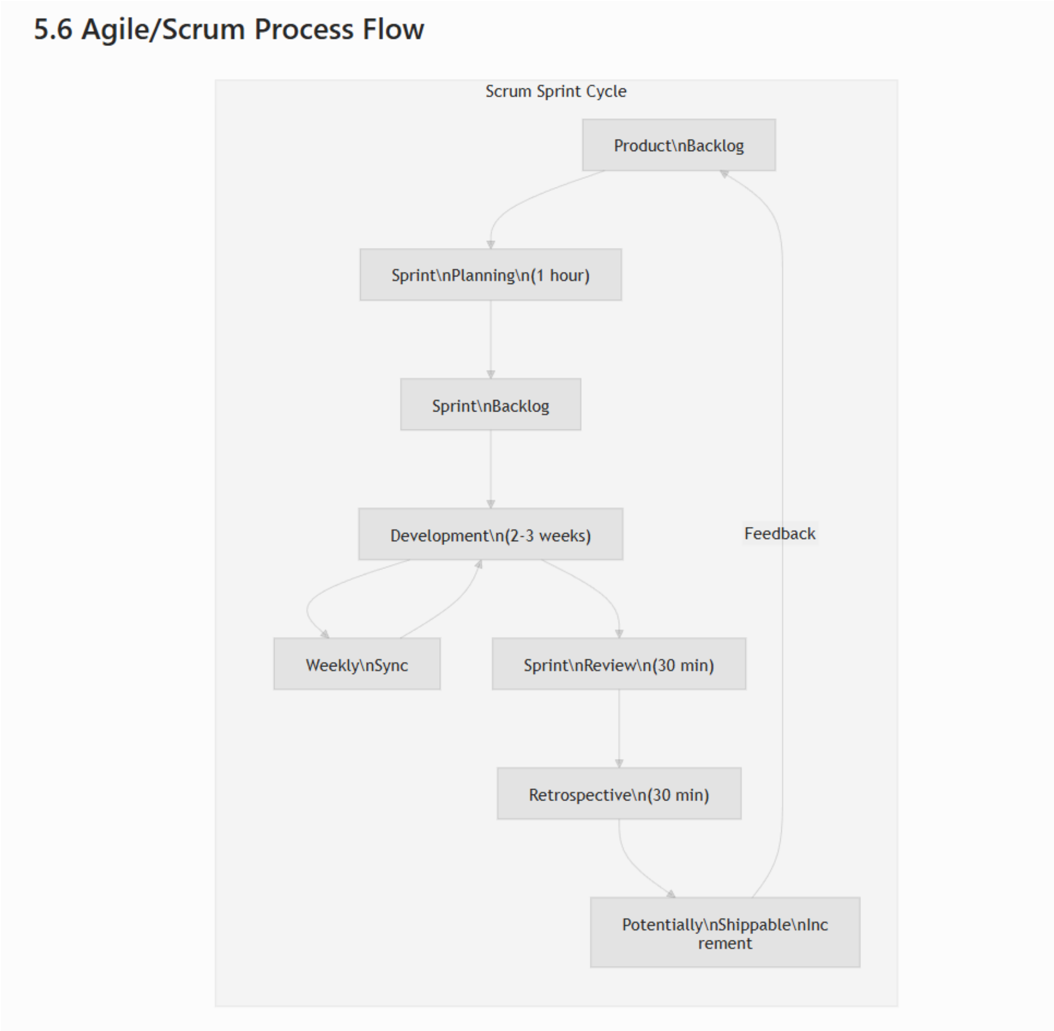
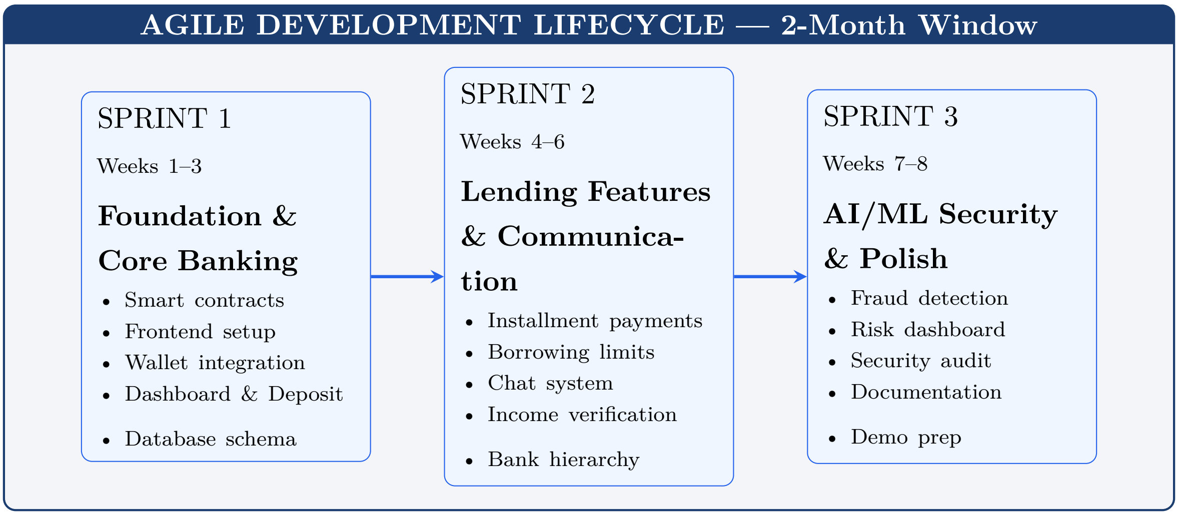


Figure 4.1: Agile/Scrum process diagram showing sprint cycles, backlog refinement, and iterative delivery within the 8-week development window.

Key parameters of our Agile implementation:

- **Sprint Duration:** 2–3 weeks per sprint (3 sprints total)
- **Team Size:** 2 developers (thesis project)
- **Weekly Sync:** Progress review, blocker identification, and planning (replaces daily standup for a 2-person team)
- **Sprint Planning:** 1 hour at sprint start
- **Sprint Review and Retrospective:** 30 minutes each at sprint end



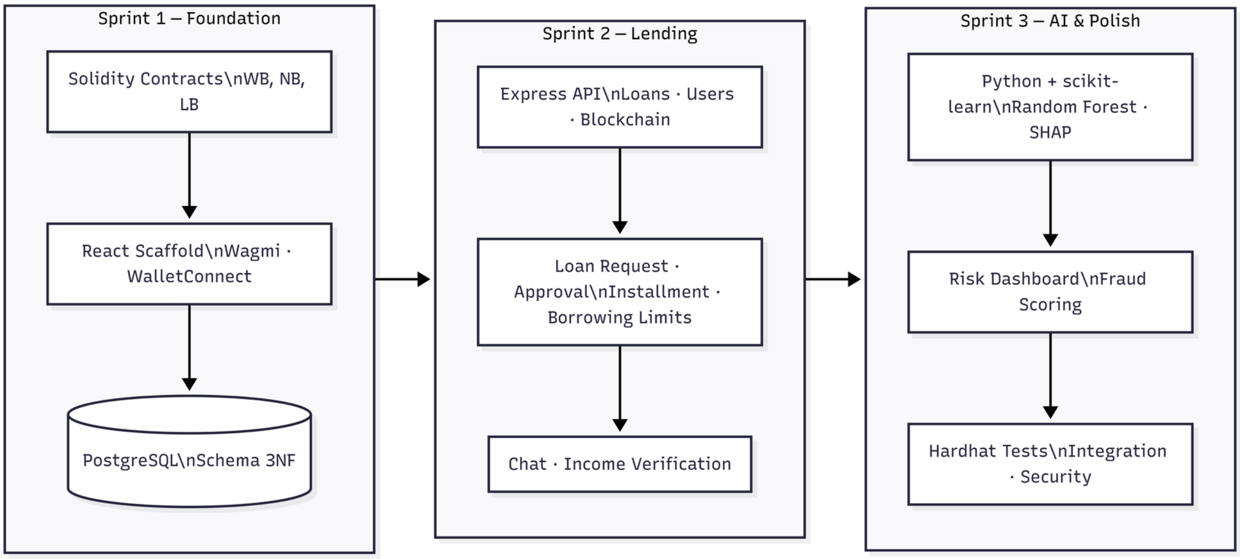


Figure 4.2: Technical development methodology: blockchain (Solidity), frontend (React, Wagmi, WalletConnect), backend (Express API, PostgreSQL), and AI/ML (Python, scikit-learn, Random Forest, SHAP) components across the three sprints.

4.2 Sprint Plan and Deliverables

4.2.1 Sprint 1: Foundation and Core Banking (Weeks 1–3)

Sprint Goal: Establish core blockchain infrastructure and hierarchical banking structure.

Table 4.1: Sprint 1 backlog: Smart contract development (21 pts).

ID	User Story	Pts
US-1.1	World Bank contract — reserve management, national bank registration	5
US-1.2	National Bank contract — borrow from WB, lend to Local Banks	5
US-1.3	Local Bank contract — borrow from NB, lend to users	5
US-1.4	Role-based access control — WB/NB/LB Admin, Bank User, Borrower	3
US-1.5	Gas cost management — initiator pays; Polygon low-fee ($0.001–0.01/tx)	3

Table 4.2: Sprint 1 backlog: Frontend foundation (13 pts) and database schema (8 pts).

ID	User Story	Pts
US-1.6	Wallet connection — MetaMask, WalletConnect, Sepolia/Mumbai	3
US-1.7	Dashboard UI — Material Design 3, responsive, blockchain-themed	5
US-1.8	Navigation and layout — AppBar, role-based menu	3
US-1.9	Blockchain visual elements — tx hash display, security badges	2
US-1.10	Database design — 15 tables, 3NF	5
US-1.11	Migration scripts and seed data	3

Sprint 1 Total: 42 story points. Deliverables: smart contracts deployed on testnet, basic frontend with wallet connection, database schema implemented, role-based access control, dashboard UI.

4.2.2 Sprint 2: Lending Features and Communication (Weeks 4–6)

Sprint Goal: Implement complete lending workflow and communication features.

Table 4.3: Sprint 2 backlog: Lending and communication features (50 pts).

ID	User Story	Pts
US-2.1	Loan request submission with blockchain transaction	5
US-2.2	Loan approval and rejection workflow with bank approver	5
US-2.3	Installment payment system with automated schedule generation	8
US-2.4	Borrowing limit engine (6-month and 1-year rolling windows)	5
US-2.5	Borrower–bank real-time chat system	5
US-2.6	Income verification document upload and review	5
US-2.7	Hierarchical bank registration (National → Local)	5
US-2.8	Bank user management and approver designation	3
US-2.9	Loan history and transaction tracking pages	5
US-2.10	QR code generation for wallet addresses	2
US-2.11	Responsive UI polish and error handling	2

4.2.3 Sprint 3: AI/ML Security and Finalization (Weeks 7–8)

Sprint Goal: Integrate AI/ML analytics, complete testing, and prepare documentation.

Table 4.4: Sprint 3 backlog: AI/ML security and finalization (38 pts).

ID	User Story	Pts
US-3.1	Random Forest fraud detection model training and deployment	8
US-3.2	SHAP-based explainability for risk assessments	5
US-3.3	Isolation Forest anomaly detection for wallet behavior	5
US-3.4	Risk dashboard with real-time AI/ML scores	5
US-3.5	AI chatbot for borrower assistance	3
US-3.6	Market data visualization page	3
US-3.7	Profile and settings management	2
US-3.8	Security audit and vulnerability assessment	3
US-3.9	Documentation and report finalization	2
US-3.10	Demo preparation and presentation	2

Figure 4.3 shows the sprint submission workflow.

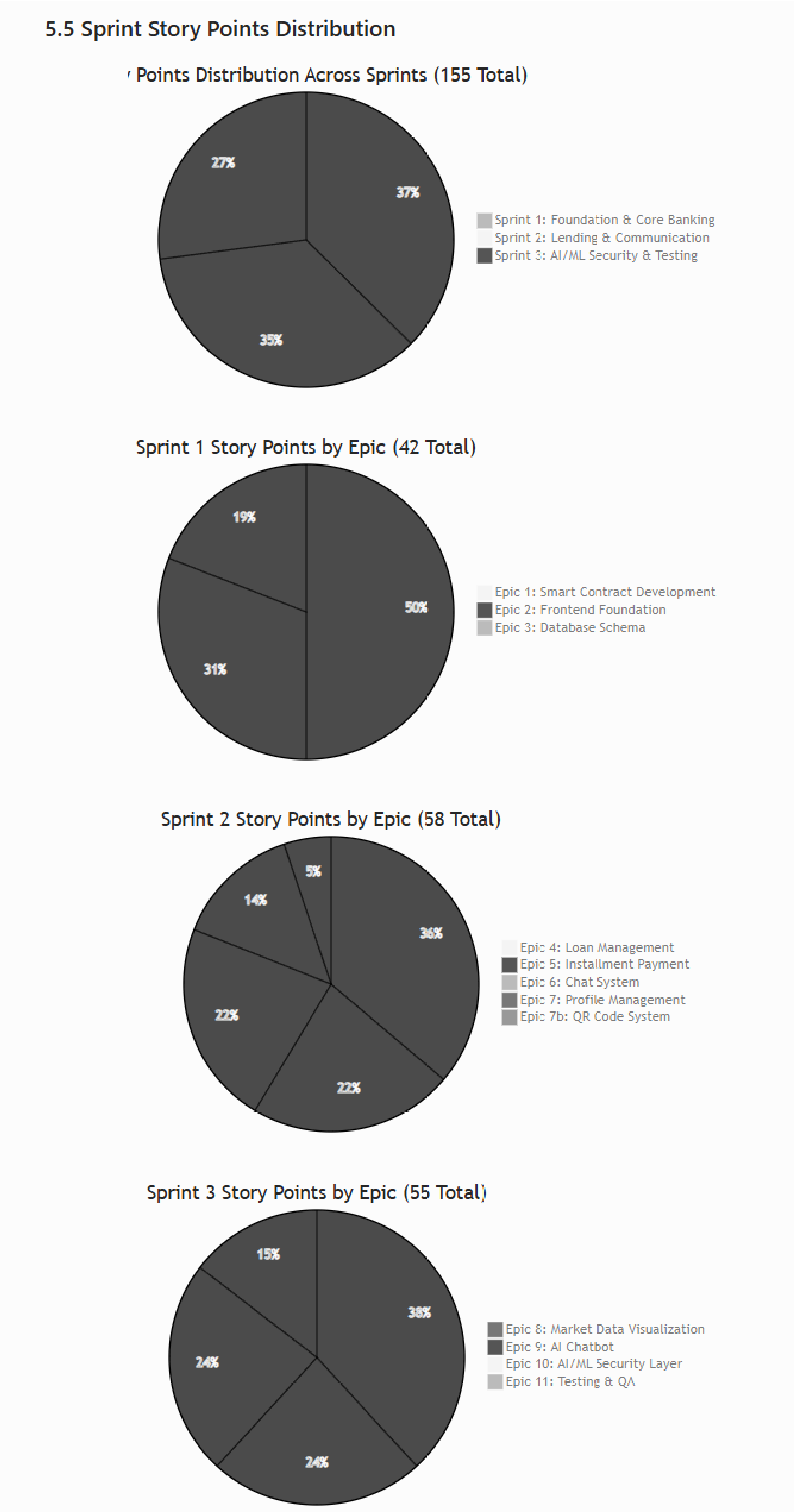


Figure 4.3: Sprint submission diagram showing the workflow from backlog refinement through development, testing, review, and delivery for each sprint cycle.

4.3 SDLC Stage Mapping

We map our Agile sprints to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.4 illustrates this mapping.

Table 4.5: SDLC stage mapping.

SDLC Stage	Project Activity	Deliverable
1. Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2. Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-5
3. Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4. Development	Sprint 1–3: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5. Testing	Hardhat unit tests (12+); integration testing; AI/ML evaluation	Test reports; model metrics
6. Deployment	Testnet deployment; frontend (Vercel); backend (Render)	Live prototype on testnet
7. Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

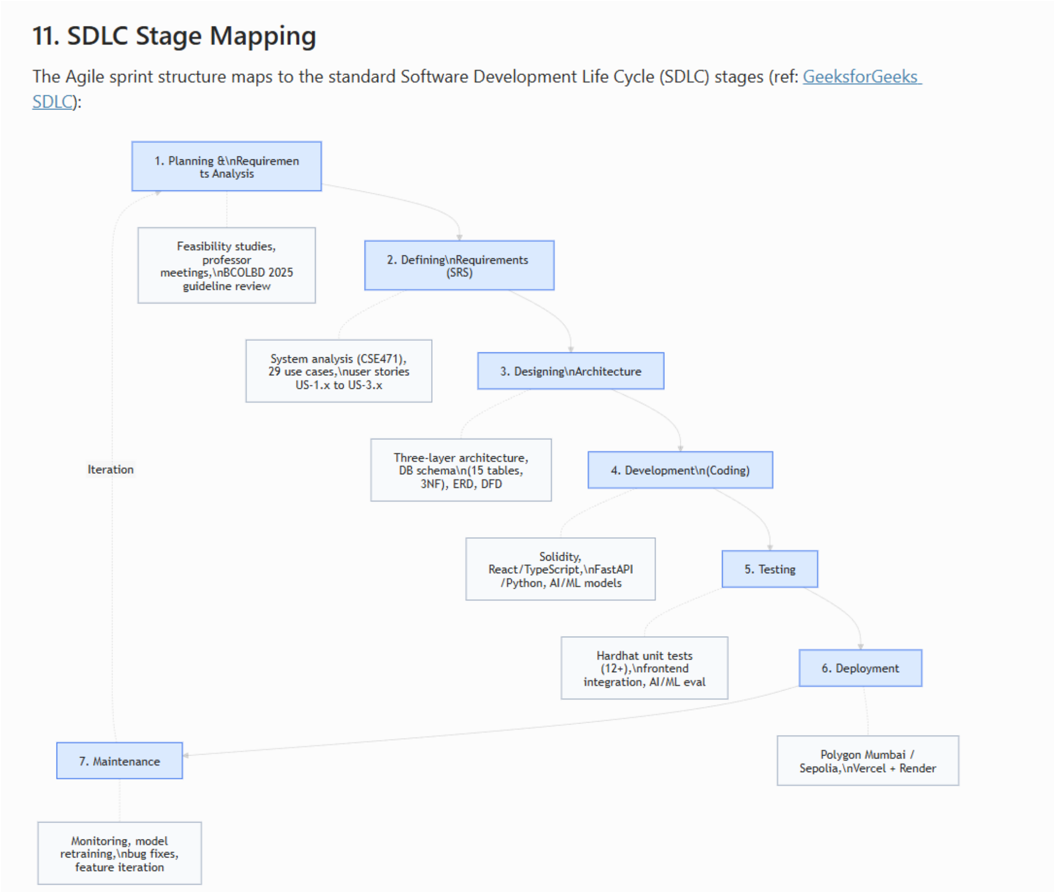


Figure 4.4: SDLC stage mapping diagram showing alignment between the seven SDLC stages and our three-sprint Agile methodology.

4.4 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints. Figure 4.5 visualizes these comparisons.

Table 4.6: Design decisions and alternatives considered.

Decision Area	**1st Choice**	**2nd Choice**	**Key Criterion**
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid w/ Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Ecosystem size, free testnets
UI Design	MUI (Material 3)	Tailwind CSS	Banking UI conventions
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, 94%+ precision
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised, no labeled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	$0 cost, publicly accessible URL

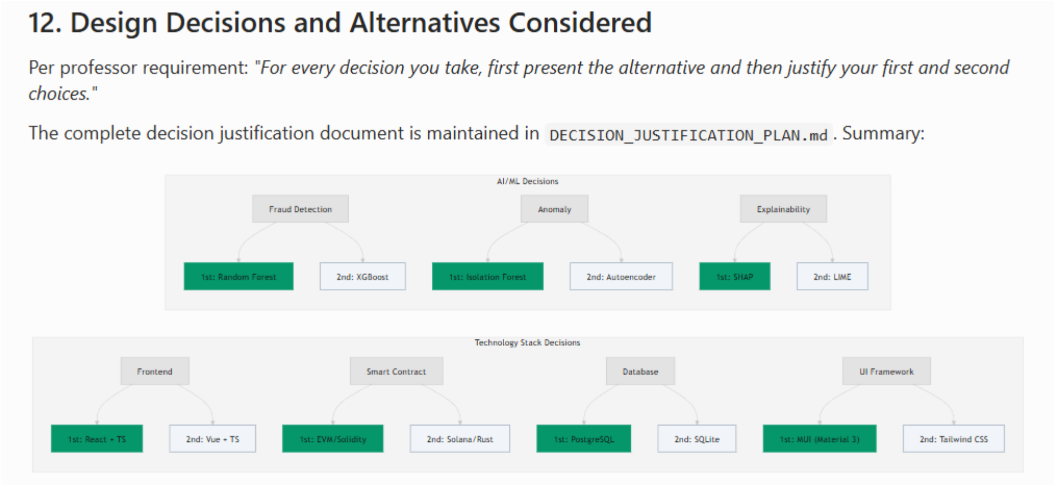


Figure 4.5: Design decisions and alternatives visualization across all major technology choices.

4.5 Design Patterns

We applied three software design patterns in our architecture:

- **Singleton:** Smart contract deployment follows the Singleton pattern—each contract (WorldBankReserve, NationalBank, LocalBank) is deployed once, and all participants interact with the same contract instance. This ensures a single source of truth for each tier's state.
- **Observer:** The event-driven architecture uses the Observer pattern. Smart contracts emit events (e.g., `LoanRequested`, `LoanApproved`, `DepositMade`) that are consumed by the off-chain event listener service, which updates the PostgreSQL database and triggers UI notifications.
- **Adapter:** The Web3 wallet integration layer implements the Adapter pattern via Wagmi and Viem, providing a unified interface across different wallet providers (MetaMask browser extension, WalletConnect mobile protocol) while adhering to the EIP-1193 standard.

4.6 Software Testing Strategy

Our testing strategy covers unit, integration, and model evaluation layers:

- **Smart contract unit tests:** 12+ test cases written in Hardhat covering reserve management, hierarchical registration, loan lifecycle, access control violations, and edge cases (zero-amount deposits, unauthorized operations).
- **Integration testing:** End-to-end workflow validation from wallet connection through loan request, approval, and installment repayment on testnet.
- **AI/ML model evaluation:** Precision, recall, F1-score, and AUC-ROC metrics

for the Random Forest fraud classifier; contamination threshold tuning for Isolation Forest; SHAP value consistency checks.

- **Frontend testing:** Manual UI testing across Chrome, Firefox, and mobile viewports for responsive layout verification.

Chapter 5

Market Analysis and Feasibility

5.1 Market Sizing

Table 5.1: Market segments.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global decentralized lending market; DeFi lending TVL has historically exceeded $50B	$50B+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures	$5–10B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance	$50–200M

Sources: *(a) TAM: DefiLlama [13] tracks DeFi TVL; lending protocols have historically exceeded $50B in total value locked; (b) SAM and SOM: Project-derived estimates based on industry segmentation [18].*

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. This represents a significant whitespace opportunity.

5.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment**—individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 5.2: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~$200 – $1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history; exclusion from banking due to documentation barriers

5.3 Partner Ecosystem

Table 5.3: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory sandbox approval; compliance oversight	Reduced enforcement cost through on-chain transparency
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced settlement friction
Payment Gateways	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic Institutions	Validation of AI/ML models; research publication	Access to anonymized datasets; collaborative opportunities
NGOs	Pilot deployment with underserved populations	Transparent, low-friction credit access for beneficiaries

5.4 Competitive Landscape

Table 5.4: Direct competition and differentiation.

Competitor Category	Examples	Our Differentiation
DeFi lending protocols	Aave, Compound, MakerDAO	Four-tier institutional model; integrated AI/ML risk layer; role-based governance
Traditional development finance	World Bank, IFC, ADB	On-chain transparency; programmable enforcement; near-instant settlement
Blockchain credit platforms	Centrifuge, Maple, Goldfinch	Combined hierarchy and AI security; designed for institutional adoption

- **Central Bank Digital Currencies (CBDCs):** Emerging CBDC frameworks may address settlement efficiency but do not provide decentralized lending capabilities. Our platform positions as a complementary **lending layer** operating on top of digital asset infrastructure.
- **Peer-to-peer lending platforms:** Predominantly fiat-denominated and centralized; the Crypto World Bank targets crypto-native and hybrid institutional use cases.

5.5 Risk Taxonomy

Table 5.5: Risk taxonomy and mitigation strategy.

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to participate	Medium	Initiate with low-barrier academic and NGO pilots
Smart contract vulnerability	Exploit in contract logic	High	OpenZeppelin primitives; formal audit (planned); pause mechanism
Regulatory adversity	Jurisdictional restrictions	Medium	Testnet-only prototype; regulatory sandbox engagement
AI/ML model degradation	Fraud detection accuracy decay	Low	Continuous retraining; human-in-the-loop; SHAP explainability

5.6 Technical Feasibility

Table 5.6: Technical feasibility assessment.

Component	Assessment	Evidence
Smart contracts	**Fully feasible**	Three contracts implemented, compiled, and tested with Hardhat (12+ passing unit tests); Solidity 0.8.20 with OpenZeppelin
Frontend DApp	**Fully feasible**	React 18 + TypeScript with all pages implemented; Wagmi and RainbowKit provide mature wallet integration
Blockchain deployment	**Fully feasible**	Polygon Mumbai and Ethereum Sepolia provide zero-cost, production-equivalent environments
AI/ML integration	**Feasible w/ constraints**	Random Forest inference achieves sub-50ms latency; SHAP explanations computable in real-time
Database backend	**Feasible**	PostgreSQL schema designed (15 tables, 3NF); FastAPI provides async REST framework

5.7 Economic Feasibility

Table 5.7: Economic feasibility — zero-cost prototype.

Cost Category	Estimate	Notes
Blockchain deployment	$0	Public testnets — no real cryptocurrency required
Frontend hosting	$0	Vercel free tier or localhost for demo
Backend hosting	$0	Render free tier or localhost
AI/ML training	$0	Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier
Development tools	$0	Hardhat, VS Code, Git — all open-source
Total prototype cost	**$0**	Entire prototype operates at zero financial cost

5.8 Revenue Projection

The following projection models annual revenue potential at full deployment scale. Calculations use a reference ETH price of **$2,000** (February 2026 market rate) and interest rate parameters defined in Section 5.8.1.

Table 5.8: Revenue projection assumptions.

Parameter	Value
Reference ETH price	\$2,000 (Feb 2026)
World Bank → National Bank rate	3% APR (wholesale inter-bank)
National Bank → Local Bank rate	5% APR (inter-bank)
Local Bank → Borrower rate	8% APR (retail lending)
Average loan term	12 months
Default rate provision	3% (conservative estimate)
Origination fee	0.25% per disbursement

Table 5.9: System-wide annual revenue summary.

Tier	Annual Revenue (ETH)	USD Equivalent
Tier 1: World Bank (1 entity)	31,525	\$63,050,000
Tier 2: National Banks (5 entities)	25,775	\$51,550,000
Tier 3: Local Banks (50 entities)	55,025	\$110,050,000
Total platform revenue	**112,325**	**\$224,650,000**
Borrower surplus generated	70,000–120,000	\$140M–240M

Sources: *ETH price from spot market data (CoinGecko); interest rate tiers aligned with Aave [15] and Compound [16] DeFi benchmarks; default rate and origination fee from conservative industry estimates.*

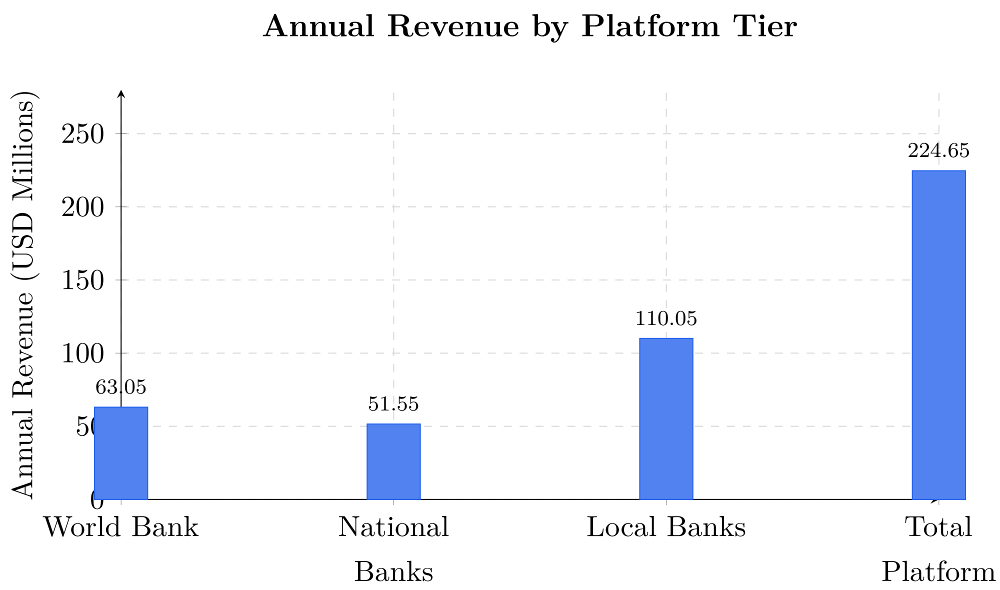


Figure 5.1: Annual revenue projection by tier (USD millions, at $2,000/ETH).

5.8.1 Transaction Economics: Interest Rates

Table 5.10: Interest rate parameters.

Parameter	Value	Benchmark
Base Annual Interest Rate	5–12% APR	Aligned with Aave/Compound variable rates
Rate Determination	Set by Local Bank approvers within World Bank-defined bounds	Configurable per-bank for local market conditions
Late Payment Penalty	2% of installment + 0.5%/week (capped at 10%)	Industry-standard late fee structure
Interest Calculation	Simple interest on outstanding principal	Transparent, borrower-friendly
Rate Transparency	All parameters stored on-chain	Publicly auditable; no hidden fees

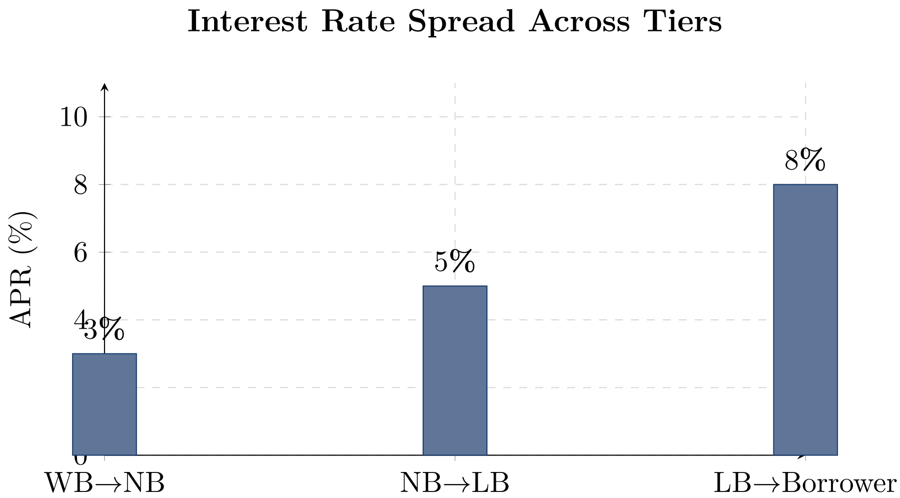


Figure 5.2: Hierarchical interest rate spread (APR) across the four-tier lending structure.

5.8.2 Global Economic Impact

Beyond direct platform revenue, the Crypto World Bank creates broader economic value:

- **Capital deployment:** \$2 billion in lending capital reaches 1,000 direct borrowers across developing economies. World Bank research [19] indicates that each dollar of institutional lending generates approximately \$2.5–3.0 in downstream economic activity, implying a potential **\$5–6 billion annual economic stimulus**.
- **Job creation:** SME lending research (IFC, 2019) [20] estimates that every \$50,000 in small business credit creates or sustains one job, suggesting support for an estimated **40,000 direct jobs**.
- **Financial inclusion:** The platform's retail-focused design provides transparent, accessible credit to underserved populations without requiring traditional banking relationships or credit histories.
- **Transaction cost reduction:** On Polygon PoS, transaction fees remain below \$0.01, representing a **60–80% cost reduction** compared to traditional correspondent banking [21].

5.8.3 Value Proposition and Go-to-Market

Table 5.11: Go-to-market phases.

Phase	Activities	Timeline
Phase 1: Validation	Competition submission (BCOLBD 2025); thesis publication; open-source release	Current
Phase 2: Pilot	Regulatory sandbox application; institutional partnership; testnet-to-mainnet migration	6–12 months
Phase 3: Production	Multi-chain deployment; full AI/ML backend; governance token launch	12–24 months

Chapter 6

Conclusion

The Crypto World Bank addresses a **complex, multi-party coordination and trust problem** in hierarchical development finance by leveraging the unique properties of blockchain technology: immutability, programmable enforcement, and cryptographic auditability. Our solution goes beyond existing DeFi lending protocols by introducing a four-tier institutional hierarchy, AI/ML-augmented risk assessment, and a governance framework that addresses network membership, business operations, and technology infrastructure.

The platform is positioned at the intersection of decentralized finance, institutional lending, and applied AI security—a combination that is presently unaddressed in both the academic literature and the commercial blockchain landscape. With a working prototype, a defined market and partnership strategy, transparent risk assessment, and a phased go-to-market plan, the Crypto World Bank represents a viable and scalable contribution to the emerging field of blockchain-based development finance.

Future work will focus on: (1) mainnet deployment with real asset testing in regulatory sandbox environments; (2) expanded AI/ML capabilities including reinforcement learning for dynamic interest rate optimization; (3) cross-chain interoperability for multi-network lending; and (4) formal security verification of all smart contract code using tools such as Slither and Mythril.

Appendix A

Technology Stack

Table A.1: Technology stack summary.

Layer	Technology	Version / Notes
Smart Contract	Solidity, OpenZeppelin	0.8.20; Ownable, ReentrancyGuard
Frontend	React, TypeScript, Vite, Material-UI	React 18; MUI v5; Material Design 3
Wallet Integration	Wagmi, RainbowKit, Viem	EIP-1193 compliant
Build and Test	Hardhat	Automated test suite; deployment scripts
Target Networks	Polygon Mumbai, Ethereum Sepolia	Public testnets; zero-cost
AI/ML (planned)	Python, FastAPI, scikit-learn, SHAP	Random Forest, Isolation Forest, SHAP

Appendix B

Smart Contract Capabilities

The platform's three smart contracts collectively support the following operations:

- **World Bank Reserve Contract:** Reserve management, deposit handling, national bank registration, fund distribution to national banks, system pause/unpause, emergency withdrawal, and global statistics.
- **National Bank Contract:** Local bank registration, borrowing from the world bank reserve, fund distribution to local banks, and network status queries.
- **Local Bank Contract:** Loan requests, loan approval and rejection, installment payments, bank user management, and approver designation.

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